Diploma in Computer Engineering

Program Outcomes (PO's)

POs are statements that describe what students are expected to know and be able to do upon graduatingfrom the program. These relate to the skills, knowledge, analytical ability, attitude, and behavior that students acquire through the program.

The POs essentially indicate what the students can do from subject-wise knowledge acquired by themduring the program. As such, POs define the professional profile of an engineering diploma graduate.

NBA has defined the following seven POs for an Engineering diploma graduate:

PO1: Basic and Discipline-specific knowledge: Apply knowledge of basic mathematics, science andengineering fundamentals and an engineering specialization to solve the engineering problems.

PO2: Problem analysis: Identify and analyse well-defined engineering problems using codified standardmethods.

PO3: Design/ development of solutions: Design solutions for well-defined technical problems and assistwith the design of systems components or processes to meet specified needs.

PO4: Engineering Tools, Experimentation, and Testing: Apply modern engineering tools and appropriatetechnique to conduct standard tests and measurements.

PO5: Engineering practices for society, sustainability and environment: Apply appropriate technology inthe context of society, sustainability, environment and ethical practices.

PO6: Project Management: Use engineering management principles individually, as a team member oras a leader to manage projects and effectively communicate about well-defined engineering activities.

PO7: Life-long learning: Ability to analyse individual needs and engage in updating in the context oftechnological changes.

DRAFT

Credit Distribution

Semester	No of Courses	Periods	Credits
Semester I	9	640	20
Semester II	9	625	20
Semester III	8	640	20
Semester IV	7	640	22
Semester V	8	605	20
Semester VI	3	660	18
		Total	**120**

DRAFT

Semester III

#	Course Category	Course Type	Code	Course Title	L-T-P	Period	Credit	End Exam
1	Program Core	Theory	1052233110	Digital Logic Design	3-0-0	45	3	Theory
2	Program Core	Practicum	1052233230	RDBMS	3-0-2	75	4	Theory
3	Program Core	Practical/Lab	1052233320	Digital Logic Design Lab	0-0-4	60	2	Practical
4	Program Core	Practicum	1052233440	C Programming	1-0-4	75	3	Practical
5	Program Core	Practicum	1052233540	Web Designing	1-0-4	75	3	Practical
6	Program Core	Practicum	1052233640	Operating Systems	1-0-2	45	2	Practical
7	Open Elective	Advanced Skill Certification	1052233760	Advanced Skills Certification - 3	1-0-2	45	2	NA
8	Humanities & Social Science	Integrated Learning Experience	1052233880	Growth Lab	0-0-2	30	1	NA
9	Audit Course	Integrated Learning Experience	1052233881-	Induction Program II	-	16	0	-
10	Audit Course	Integrated Learning Experience	-1052233882-	I&E/ Club Activity/ Community Initiatives	-	16	0	-
11	Audit Course	Integrated Learning Experience	1052233883	Shop floor Immersion	-	8	0	-
12	Audit Course	Integrated Learning Experience	-1052233884	Student-Led Initiative	-	6	0	-
13	Audit Course	Integrated Learning Experience	-1052233885-	Emerging Technology Seminars	-	30	0	-

DRAFT

14	Audit Course	Integrated Learning Experience	1052233886-	Health & Wellness	-	24	0	-
					Test & Revisions	**75**		
					Library	**15**		
					Total Periods	**640**	**20**	

Semester IV

#	Course Category	Course Type	Code	Course Title	L-T-P	Period	Credit	End Exam
1	Program Core	Theory	1052234110	Computer Networks and Security	3-0-0	45	3	Theory
2	Program Core	Practicum	1052234230	Data Structures Using Python	3-0-2	75	4	Theory
3	Program Core	Practicum	1052234340	Java Programming	2-0-4	90	4	Practical
4	Program Core	Practicum	1052234440	Python Programming	1-0-4	75	3	Practical
5	Program Core	Practicum	1052234540	E-Publishing Tools	1-0-4	75	3	Practical
6	Program Core	Project/Internship	1052234640	Scripting Languages	0-0-6	90	3	Practical
7	Open Elective	Advanced Skill Certification	1052234760	Advanced Skills Certification - 4	1-0-2	45	2	NA
8	Audit Course	Integrated Learning Experience	-1052234882-	I&E/ Club Activity/ Community Initiatives		30	0	
9	Audit Course	Integrated Learning Experience	1052234883	Shop floor Immersion	-	8	0	-
10	Audit Course	Integrated Learning Experience	-1052234884	Student-Led Initiative	-	24	0	-
11	Audit Course	Integrated Learning	-1052234885-	Emerging Technology	-	8	0	-

DRAFT

		Experience		Seminars				
12	Audit Course	Integrated Learning Experience	1052234886-	Health & Wellness	-	30	0	-
13	Audit Course	Integrated Learning Experience	-1052234887-	Special Interest Groups (Placement Training)	-	30	0	-
					Test & Revisions	45		
					Library	15		
					Total Periods	**640**	**22**	

Semester V

#	Course Category	Course Type	Code	Course Title	L-T-P	Period	Credit	End Exam
1	Program Core	Practicum	1052235130	Cloud Computing	2-0-2	60	3	Theory
2	Program Elective	Theory		Elective-1	3-0-0	45	3	Theory
3	Program Core	Practical/Lab	1052235320	Internet of Things & Digital Twins	0-0-4	60	2	Practical
4	Program Core	Practicum	1052235440	Computer Hardware and Networking	1-0-4	75	3	Practical
5	Program Core	Practicum		Elective-2	1-0-4	75	3	Practical
6	Humanities & SocialScience	Practicum	xxxx235640	Innovation & Startup	1-0-3	60	2	Project
7	Project / Internship	Project/Internship	1052235773	Industrial Training * [Summer Vacation - 90 Hours]	-	-	2	Project
8	Open Elective	Advanced Skill Certification	1052235860	Advanced Skills Certification - 5	1-0-2	45	2	NA
9	Audit Course	Integrated Learning Experience	-1052234882-	I&E/ Club Activity/ Community Initiatives		30	0	

DRAFT

10	Audit Course	Integrated Learning Experience	1052234883	Shop floor Immersion	-	8	0	-
11	Audit Course	Integrated Learning Experience	-1052234884	Student-Led Initiative	-	24	0	-
12	Audit Course	Integrated Learning Experience	- 1052234885	Emerging Technology Seminars	-	8	0	-
13	Audit Course	Integrated Learning Experience	1052234886	Health & Wellness	-	30	0	-
14	Audit Course	Integrated Learning Experience	- 1052234887	Special Interest Groups	-	30	0	-
					Test & Revisions	**60**		
					Library	**15**		
					Total Periods	**605**	**20**	

Elective 1

#	Course Category	Course Type	Course Code	Course Title	L-T-P	Period	Credit	End Exam
1	Program Elective	Theory	1052235211	Machine Learning	3-0-0	45	3	Theory
2	Program Elective	Theory	1052235212	Data Warehousing and Data Mining	3-0-0	45	3	Theory
3	Program Elective	Theory	1052235213	Ethical Hacking	3-0-0	45	3	Theory
4	Program Elective	Theory	1052235214	Agile Product Development	3-0-0	45	3	Theory
5	Program Elective	Theory	1052235215	Artificial Intelligence	3-0-0	45	3	Theory

DRAFT

Elective 2

#	Course Category	Course Type	Course Code	Course Title	L-T-P	Period	Credit	End Exam
1	Program Elective	Practicum	1052235541	Data Analytics	1-0-4	75	3	Practical
2	Program Elective	Practicum	1052235542	Mobile Computing	1-0-4	75	3	Practical
3	Program Elective	Practicum	1052235543	Component Based Technologies	1-0-4	75	3	Practical
4	Program Elective	Practicum	1052235544	Multimedia Systems	1-0-4	75	3	Practical
5	Program Elective	Practicum	1052235545	Full Stack Developer	1-0-4	75	3	Practical
6	Program Elective	Practicum	1052235546	Robotic Process Automation	1-0-4	75	3	Practical

DRAFT

Semester VI

#	Course Category	Course Type	Course Title	L-T-P	Period	Credit	End Exam
1	Open Elective	Theory	Electives-3 (Pathway)	3-0-0	45	3	Theory
2	Open Elective	Practicum	Elective-4 (Specialisation)	1-0-4	75	3	Practical
3	Project / Internship	Project / Internship	In-house Project / Internship / Fellowship	-	540	12	Project
			Total Periods		**660**	**18**	

Elective 3 (Pathway)

#	Course Category	Course Type	Course Code	Course Title	L-T-P	Period	Credit	End Exam
1	Elective | Higher Education	Theory	XXXX236111	Advanced Engineering Mathematics	3-0-0	45	3	Theory
2	Elective | Entrepreneurship	Theory	XXXX236112	Entrepreneurship	3-0-0	45	3	Theory
3	Elective | Technocrats	Theory	XXXX236113	Project Management	3-0-0	45	3	Theory
4	Elective | Technocrats	Theory	XXXX236114	Finance Fundamentals	3-0-0	45	3	Theory
5	Elective | Technologists	Theory	1052236115	5G Technology	3-0-0	45	3	Theory
6	Elective | Technologists	Theory	1052236116	DevOps	3-0-0	45	3	Theory

Elective 4 (Specialisation)

#	Course Category	Course Type	Course Code	Course Title	L-T-P	Period	Credit	End Exam
1	Elective	Practicum	1052236241	Data Science	1-0-4	75	3	Practical
2	Elective	Practicum	1052236242	Cloud Platform	1-0-4	75	3	Practical
3	Elective	Practicum	1052236243	Data Visualization	1-0-4	75	3	Practical
4	Elective	Practicum	1052236244	Advance DBMS	1-0-4	75	3	Practical
5	Elective	Practicum	1052236245	Mobile Application Development	1-0-4	75	3	Practical

DRAFT

6	Elective	Practicum	1052236246	UI & UX Design	1-0-4	75	3	Practical

Project / Internship

#	Course Category	Course Type	Course Code	Course Title	L-T-P	Period	Credit	End Exam
1	Project / Internship	Project / Internship	1052236351	Internship	-	540	12	Project
2	Project / Internship	Project / Internship	1052236353	Fellowship	-	540	12	Project
3	Project / Internship	Project / Internship	1052236374	In-house Project	-	540	12	Project

DRAFT

1052233110	DIGITAL LOGIC DESIGN	L	T	P	C
Theory		3	0	0	3

Introduction:

This subject introduces students to the fundamental concepts and techniques for designing and analysing digital circuits, laying the groundwork for understanding and creating digital technologies.

Course Objectives:

The objective of this course is to enable the students to

- Provide comprehensive understanding of digital systems and their fundamental components, applications.
- Simplify and optimize digital logic circuits while gaining practical insights into its applications through Boolean algebra.
- Learn how to design sequential logic circuits using various components and techniques.
- Learn about digital sensor interfaces and their role in digital systems.
- Understand the principles and operation of various Analog-to-Digital Converters (ADC) and Digital-to-Analog Converters (DAC).
- Gain knowledge of different types of memory and their characteristics.
- Understand Programmable Logic Devices (PLDs) and their applications in digital system design.
- Analyze the societal impact of Digital Integrated Circuits (ICs) and their role in various industries and technologies.

This initial course offers students a gateway into the realm of digital electronics.

Course Outcomes:

On successful completion of this course, the student will be able to

CO1: Understanding digital systems fundamentals comprehensively.

CO2: Explore Boolean algebra fundamentals and practical applications.

CO3: Utilize sequential logic principles to create designs.

CO4: Design digital sensor interfaces and ADC/DAC converters.

CO5: Analyse memory and PLDs.

DRAFT

Pre-requisites: Nil

CO/PO Mapping:

CO / PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2	2	2	1	-	-	1
CO2	2	2	2	2	-	-	1
CO3	2	2	2	2	-	-	1
CO4	2	2	2	1	-	-	1
CO5	2	2	2	2	1	1	1

Legend: 3-High Correlation, 2-Medium Correlation, 1-Low Correlation

Instructional Strategy:

Real-world Applications: Integrate real-world examples and applications of digital logic design, such as binary arithmetic in computer architecture, digital communication systems, and control systems. Showing practical applications helps students understand the relevance of the subject.

Interactive Lectures: Conduct interactive lectures with demonstrations, multimedia presentations, and interactive whiteboards to illustrate abstract concepts effectively. Encourage student participation through discussions, questions, and problem-solving exercises.

Case Studies: Present case studies of real-world digital systems, highlighting design challenges, solutions, and outcomes. Analysing case studies helps students understand the design process, constraints, and considerations involved in digital logic design.

Use of Visual Aids: Utilize visual aids such as diagrams, charts, and animations to clarify complex concepts like Boolean algebra, logic gates, and sequential logic circuits. Visual representations help reinforce learning and improve comprehension.

Flipped Classroom Approach: Implement a flipped classroom model where students review lecture materials and resources independently before class and use class time for hands-on activities, problem-solving, and discussions. This approach encourages active learning and fosters deeper understanding.

Formative Assessment: Use formative assessment techniques such as quizzes, concept mapping, and in-class exercises to gauge student understanding and provide timely feedback. Adjust teaching strategies based on assessment results to address areas of

DRAFT

difficulty.

Self-directed Learning Resources: Provide self-directed learning resources such as textbooks, online tutorials, and supplementary materials to accommodate diverse learning styles and allow students to explore topics at their own pace.

Assessment Methodology:

	Continuous Assessment (40 marks)			End Semester Examination (60 marks)
	CA1	CA2	CA3	
Mode	Written Test (Unit I & II)	Written Test (Unit III & IV)	Model Theory Exam	Theory Exam
Duration	2 Hours	2 Hours	3 Hours	3 Hours
Exam Marks	50	50	100	100
Converted to	20	20	20	60
Final Marks	40			60

Note:

- CA1 and CA2 Assessment tests should be conducted.CA1 covers unit 1 and unit 2, CA2 covers unit 3 and unit 4. Best of one will be considered for 20 Marks for internal assessment.
- CA3 Model examination should be conducted as per the end semester question pattern after completion of all the portions. The marks awarded shall be converted to 20 Marks.

DRAFT

1052233110	DIGITAL LOGIC DESIGN	L	T	P	C
Theory		3	0	0	3

Unit I	Foundations of Digital Logic	
Introduction: digital systems and their importance, Number systems: binary, octal, hexadecimal number, binary arithmetic: 1's and 2's complements arithmetic, binary conversion, binary code standards: ASCII, BCD, Graycode, and Unicode. Logic Gates: AND, OR, NOT, NAND, NOR and Ex-OR operations and logic diagram, realization of gates using universal gates		8
Unit II	**Combinational Logic Design**	
Basic concepts of Boolean algebra: Laws, theorems, De-Morgan's theorem, Standard representation of logic functions, K-map representation (upto 3 variable), simplification of logic functions using K-map. Applications: Adders, Subtractors, digital comparator, parity checker / generator, Multiplexer(MUX)/ data selector, De-Multiplexer(D-UX)/decoder, operation of a 4-to-10-line decoder, operation of an 8-to-3-line encoder, priority encoder.		9
Unit III	**Sequential Logic Design**	
Flip Flops: SR - JK-T and D types flip flops, concept of Edge Triggering and Level Triggering, applications of flip flops. Registers: shift registers, serial to serial converter, serial to parallel converter, parallel to serial converter, applications of shift registers. Counters: Asynchronous up counters, decade ripple counter, Synchronous down counters, applications of counters.		9
Unit IV	**Digital Interfacing , ADC and DAC**	
Digital Interfacing: Introduction, sensors and their types, TTL and CMOS interface - switch, LED, relay, motor and solenoid. A/D converters: successive approximation A/D converter, dual slope A/D converter, D/A converters: weighted resistor/converter, R-2RLadder D/A converter.		9
Unit V	**Memories, Programmable Logic Devices**	

DRAFT

Memory: Definition- i) memory read ii) memory write iii) access time iv) memory capacity v) address lines vi) word length, Different types of ROM & RAM, Memory accessing, processing, hierarchy and management, difference between Flash ROM and NVRAM, operation of pen drive, SD card and solid state hard disk. PLDs: Difference between fixed logic and programmable logic, PLA and PAL-architecture, and implementation of simple Boolean equations.	10
TOTAL PERIODS	**45**

Suggested List of Students Activity:

The following student activities or similar activities can be assigned

1. Collect the information about the different types of display devices used in digital circuits and carry out a seminar.
2. Prepare a note on E-waste and disposal of PCBs and ICs, carry out a seminar.
3. Organize a series of problem-solving sessions where students work on Boolean algebra problems and logic circuit design tasks. Evaluate students based on their ability to manipulate Boolean expressions, simplify logic circuits, and design practical solutions. Assess their problem-solving skills, logical reasoning, and the correctness of their solutions
4. Provide industrial sensor datasheets or specifications for commonly used sensors in manufacturing processes (e.g., temperature sensors, pressure sensors, proximity sensors).
5. Ask students to design and implement digital sensor interfaces or ADC/DAC circuits to interface with these sensors and provide digital or analog measurements.
6. Provide specifications or requirements for industrial control tasks, such as sequence control, timing control, or data processing tasks. Ask students to design, simulate, and implement the sequential logic circuits using PLC programming software or microcontroller platforms commonly used in industrial automation.
7. Assign a group project where diploma students analyse the memory technologies, programmable logic devices (PLDs), and their societal impact in industrial applications. Focus on memory technologies commonly used in industrial control systems (e.g., EEPROM, flash memory) and PLDs used for logic control and customization in industrial automation.

Execution Notes:

DRAFT

- Maximum of 3 students in each batch for student activity
- Above activities may be distributed among different batches; Any one activity among 1 to 5 or any similar activities per batch may be assigned by the teacher based on interest of the students.

Text Books:

1. "Digital logic and Computer design", M. M. Mano, Pearson Education India, 2016.
2. ”Introduction to Digital Logic Design”, J P. Hayes, Addison-Wesley Publishing Co.
3. "Digital Systems: Principles and Applications" by Ronald J.Tocci, Neal S. Widmer, and Gregory L. Moss.

Reference Books:

1. "Digital Electronics: Principles, Devices, and Applications" by Anil K. Maini
2. “Digital Principles and Applications” Malvino & Leach, McGraw-Hill Education.
3. "Modern Digital Electronics", R. P. Jain, McGraw Hill Education, 2009.
4. ”Digital Principles and Design”, John F Wakerly, Donald D.Givone, Tata McGraw Hill, Edition.
5. ”Digital Design Principles and Practices by Wakerly, John 5/e, Pearson Education 2018.
6. "Fundamentals of Digital Circuits", A. Kumar, Prentice Hall India, 2016
7. ” Fundamentals of logic design, Charles H. Roth, Jr.Cengage Learning, New Delhi.
8. “Digital Design and Computer Architecture: ARM Edition, Sarah Harris and David Harris, 2015.

Web-based/Online Resources:

https://nptel.ac.in/

https://www.electronicsforu.com/

http://www.freebyte.com/electronics/

https://www.edaplayground.com/

https://www.circuitlogix.com

http://www.vlab.co.in

http://www.allaboutcircuits.com

Continuous Assessment Pattern (CA1 & CA2)

Theory (50)

Allocation of Marks for CA1 & CA2 Theory Tests

DRAFT

Section	Description	Marks	
Part – A	11 questions to be answered out of 12	11 x 2 marks	22
Part – B	4 questions to be answered out of 6	4 x 7 marks	28
TOTAL			**50**

1052233230	**RDBMS**	L	T	P	C
Practicum		3	0	2	4

Introduction

All modern database management systems like SQL, MS SQL Server, IBM DB2, ORACLE, My-SQL, and Microsoft Access are based on RDBMS. It is called Relational Database Management System (RDBMS) because it is based on the relational model introduced by E.F. Codd. A relational database is the most commonly used database. Due to a collection of an organized set of tables, data can be accessed easily in RDBMS.

Course Objectives

The objective of this course is to enable the student to

- To know the fundamentals of DBMS
- To share of data and speedy forming of new applications, restrict repetition or redundancy of data
- To avoid data inconsistencies providing better integrity
- To familiarize all the possible operations of data in the database
- To familiarize programming skills for all the operations in database

Course Outcomes

After successful completion of this course, the students should be able to

CO1:Describe fundamentals of DBMS, installation and working with admin

CO2:Apply SQL commands to create, insert, retrieve, update, delete data from the Relational databases.

CO3: Describe MySQL programming constructs, control statements and subprograms.

CO4: Describe how to Tune MySQL performance

DRAFT

CO5: Apply cursors, triggers and Exception handling concepts

Pre-requisites: Nil

CO/PO Mapping

CO / PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	2	2	3	2	---	---
CO2	3	2	3	2	3	---	---
CO3	2	3	2	2	2	---	---
CO4	2	2	3	3	3	---	---
CO5	3	3	3	2	2	---	---

Legend: 3-High Correlation, 2-Medium Correlation, 1-Low Correlation

Instructional Strategy

- Engage and Motivate: Instructors should actively engage students to boost their learning confidence.
- Real-World Relevance: Incorporate relatable, real-life examples and engineering applications to help students understand and appreciate course concepts.
- Interactive Learning: Utilize demonstrations and plan interactive student activities for an engaging learning experience.
- Application-Based Learning: Employ a theory-demonstrate-practice-activity strategy throughout the course to ensure outcome-driven learning and employability.
- Encourage Critical Analysis: Foster an environment where students can honestly assess experiment outcomes and analyze potential sources of error in case of discrepancies.

DRAFT

Assessment Methodology

	Continuous Assessment (40 marks)					End Semester Examination (60 marks)
	CA1		CA2		CA3	
Mode	Written Test (First 50% of Theory)	Practical Test (First 50% of Exercise)	Written Test (Next 50% of Theory)	Practical Test (Next 50% of Exercise)	Model Theory Exam	Theory Exam
Duration	2 hours	2 hours	2 hours	2 hours	3 hours	3 hours
Exam Marks	50	50	50	50	100	100
Converted to	20		20		10	60
Practical log book / Observation Note book with bonafide	10					
Final Marks	40					60

Note:

DRAFT

- Continuous Assessment tests (CA1 and CA2) should be conducted for both Theory and Practical portions. **Best of one** will be considered for 20 marks for internal assessment.
- Continuous Assessment Model Theory examination (CA3) should be conducted as per the end semester question pattern after completion of all the portions. The marks awarded shall be converted to 10 Marks.
- Practical log book / Observation note book with bonafide carries 10 Marks.

DRAFT

1052233230	RDBMS	L	T	P	C
Practicum		3	0	2	4

Unit I	RELATIONAL DATA MODEL & MYSQL ADMINISTRATION	
	Relational Data Model Introduction to database concepts- CODD's Rules – Components of DBMS – Table Structure – Keys : types of Keys – Data Constraints and types of Constraints - difference between SQL and MySQL **MySQL Installation** Install ,configure and test MySQL server on Microsoft Windows **Working with MySQL Admin** Creating , selecting and describing database – show command – backing up databases	9
	Ex No 1: Install, configure and connect to MySQL server and MySQL workbench in Windows **Ex No 2**: Write a query to create a database, show and backing up databases	6
Unit II	**INTERACTIVE MYSQL**	
	Introduction to MYSQL MySQL data types – Data Definition Commands – Data Manipulation Commands – Data Retrieval Commands **MySQL operators and Expressions**	9

Types of operators – Arithmetic, Comparison and Logical Operators – Pattern Matching- Import and Export Data **Built – in Functions** Single Row functions – Aggregate functions – conversion functions	
Ex No 3: create a database named 'college' and create a table for student and employee with the fields as you like. **Ex No 4:** create a table 'student' with marks field for 10 students. Apply built in functions to do calculations.	6
Unit III	***FLOW CONTROL IN MYSQL***
Flow Control IF(), IF NULL(),CASE ,LOOP,LEAVE ,ITERATE , REPEAT,WHILE **Querying the table** Selecting rows using where, order by, group by and Having clauses-Sub-queries-correlated sub-queries **Views** Introduction – Advantages of views – creating , updating and deleting views	9
Ex No 5: create a table 'bank' and apply flow control statements to do some transactions **Ex No 6:** create a table ' library' with proper fields and create another table ' library1' and insert rows from library using views	6
UNIT IV	***MYSQL PERORMANCE TUNING***
Indexes and Sequences Index Types – simple and compound – Sequences : creating , altering and dropping sequences **Joins & Unions** Joins- definition-types of joins : natural join,inner join,self join,outer join. Union types : union , Union All, Union Distinct- order by and limit handling **User and transaction management** Creating – deleting – renaming users grant and and revoke commands – transaction command : commit , rollback and save points.	9
Ex No 7: create a table named 'student' with sequences **Ex No 8**: create any two tables with common column name and perform join and union	6

DRAFT

UNIT V	*STORED PROGRAM CONCEPTS & DEVELOPMENT*	
MySQL Procedures & Functions Creating –executing and deleting stored procedures – creating – executing and deleting stored functions -advantages **MySQL Trigger & Cursor** Use of trigger – creating trigger – types of triggers Cursor: creation and deletion		9
Ex No 9 : Create a stored procedure to get employee details from employee table **Ex No 10:** Create a program for trigger and cursor		6
TOTAL HOURS		75

REFERENCE BOOKS

1. Querying MySQL: Make your MySQL database analytics accessible with SQL operations, data extraction, and custom queries, by Adam Aspin
2. MySQL: The Complete Reference (Osborne Complete Reference Series) by VikramVaswani | 16 January 2004
3. MySQL Pocket Reference 2e (Pocket Reference (O'Reilly)), by George Reese | 7 August 2007
4. High Performance MySQL: Optimization, Backups,Replication, Third Edition by Baron Schwartz , Peter Zaitsev, et al. | 15 May 2012
5. https://www.w3schools.com/mysql/
6. https://www.mysqltutorial.org/
7. https://www.javatpoint.com/mysql-tutorial
8. https://www.guru99.com/mysql-tutorial.html

Suggested List of Students Activity

- Presentation/Seminars by students on any recent technological developments based on the course.
- Periodic class/online quizzes conducted based on the course.
- Blended learning activities to explore the recent trends and developments in the field.

Equipment / Facilities required to conduct the Practical Portion

1. **Hardware Requirement:**
    - Desktop Computers / Laptop
    - Printer

2. **Software Requirement:**

- Windows / Linux Operating System
- https://dev.mysql.com/downloads/installer/

Continuous Assessment Pattern (CA1 & CA2)

Theory (50)

Allocation of Marks for CA1 & CA2 Theory Tests

Section	Description	Marks	
Part – A	11 questions to be answered out of 12	11 x 2 marks	22
Part – B	4 questions to be answered out of 6	4 x 7 marks	28
TOTAL			**50**

Practical (50)

Allocation of Marks for CA1 & CA2 Practical Tests

Section	Description	Marks
1	Aim & Procedure	35
2	Execution and Result	15
TOTAL		**50**

DRAFT

1052233320	DIGITAL LOGIC DESIGN LAB	L	T	P	C
PRACTICAL		0	0	4	2

Rationale

This course emphasizes practical application alongside theoretical knowledge, covering digital logic devices and circuits, hands-on experiments, understanding IC specifications, and application through simulation exercises and mini-projects, fostering critical thinking and problem-solving abilities.

Course Objectives

The objective of this course is to

1. To familiarize students with digital logic devices and circuits, including logic gates, flip-flops, multiplexers, and comparators.
2. To enable students to understand the specifications and characteristics of digital ICs through the study of datasheets.
3. To provide hands-on experience in implementing and verifying truth tables using TTL and CMOS gates.
4. To introduce students to digital interface circuits and their applications in controlling external devices like LEDs and DC motors.
5. To explore digital-to-analog and analog-to-digital conversion principles using DAC and ADC ICs.
6. To develop students' skills in designing and troubleshooting digital circuits through simulation and practical experimentation.

Course Outcomes

After successful completion of this course, the students should be able to

CO1: Test the truth tables of logic gates. Build combinational logic circuits and validate their truth tables

CO2: Construct sequential logic circuits and verify their truth tables.

CO3: Design digital interfaces for LED, motor; DAC/ADC connection.

CO4: Utilize PAL, PLA for Boolean; Perform SRAM operations.

CO5: Simulate digital logic circuits using P-Spice or similar software.

Pre-requisites

DRAFT

- Students should have knowledge of basic logic gate operations, Boolean algebra, and digital circuit design principles.
- Additionally, having hands-on experience with electronic components, bread boarding, and using simulation software tools would be beneficial.

CO/PO Mapping

CO / PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2	2	2	2	-	-	1
CO2	2	2	1	2	-	-	1
CO3	2	2	1	2	-	-	1
CO4	2	1	2	2	1	1	1
CO5	2	2	2	2	1	1	1
CO6	2	1	1	2	-	2	1

Legend: 3-High Correlation, 2-Medium Correlation, 1-Low Correlation

Instructional Strategy

- It is advised that teachers make the learning experience more engaging by introducing innovative and interesting ways of teaching. The teachers need to expose the students to material in multiple modes help them learn it faster and retain it longer.
- **Use of Visual Aids:** Utilize visual aids such as diagrams, charts, and animations to clarify complex concepts like Boolean algebra, logic gates, and sequential logic circuits. Visual representations help reinforce learning and improve comprehension.
- **Hands-on Labs and Simulations:** Provide hands-on experience with logic gates, flip-flops, and other digital components through laboratory sessions. Additionally, utilize digital logic simulation software to allow students to design and simulate circuits, providing a practical understanding of concepts
- **Group Projects:** Assign group projects that require students to design and implement digital circuits to solve specific problems or tasks. This encourages teamwork, enhances problem-solving skills, and reinforces learning through practical application.

DRAFT

- **Guest Lectures and Industry Visits:** Invite guest speakers from industry or academia to share insights into digital logic design applications, emerging technologies, and career opportunities. Organize visits to relevant industries or laboratories to expose students to real-world digital design practices.

DRAFT

Assessment Methodology

	Continuous Assessment (40 marks)			End Semester Examination (60 marks)
	CA1	CA2	CA3	
Mode	Practical Test First 50% Exercises	Practical Test Next 50% Exercises	Model Practical Examination	Practical Exam
Duration	2 hours	2 hours	3 hours	3 hours
Exam Marks	50	50	100	100
Converted to	20	20	10	60
Practical log book / Observation Notebook with bonafide.	10			
Final Marks	40			60

Note:

- Both CA1 and CA2 Assessment test should be conducted. Best of CA1 and CA2 should be considered for the internal assessment of 20 Marks.
- Continuous Assessment Model Practical examination (CA3) should be conducted as per the end semester question pattern after completion of all the portions. The marks awarded shall be converted to 10 Marks.
- The student should submit the duly verified practical log/observation book during CA3..
- Practical logbook / Observation Notebook with Bonafide Carries 10 Marks.

DRAFT

1052233320	DIGITAL LOGIC DESIGN	L	T	P	C
Practical		0	0	4	2

Part - A	Digital Logic devices &Circuits : Construct & Verify	
Nomenclature of Digital ICs, study of the datasheet, concept of Vcc and ground – precautions of using Digital ICs.		
Ex.No	**Name of the Experiment**	
1	Verify the truth tables of (i) Logic Gates (any 3) (ii) Flip Flops (any 3).	30
2	Implement 2 input Boolean logic function using logic gates in (i) POS form (ii). SOP form.	
3	Verify the truth tables of (i) Multiplexer IC74151 (ii) Encoder IC74148.	
4	Implement and verify the truth table using TTL/CMOS gates. ((i) BCD to 7 segments Decoder (ii) 4-bit Ripple Counter.	
5	Verify the function of (i) 4-bit magnitude comparator 7485 IC (ii)Shift registers (ICs like 7495, 74194 etc.).	
Part - B	**Digital Interface, ADC/DAC and Memory : Demonstrate & Simulate**	
Introduction to Pspice Simulation tool, designing procedure for simple circuits.		
Ex.No	**Name of the Experiment**	
6	Design a digital interface circuit to control the illumination of an LED and control the speed and direction of a DC motor using TTL/CMOS ICs.	30
7	Connect DAC0808 IC to convert digital to analog and ADC0804 IC to convert analog to digital. Show analog output via oscilloscope and digital via LEDs.	

DRAFT

8	Implement the Boolean expression F = A'B'C' + A'BC + AB'C + ABC' using (i) PAL16R8 IC (ii) PLA22V10 IC.	
9	Perform SRAM Operations through Reading, Writing using IC6264.	
10	**Simulate** (i) Full Adder Circuits Using ICs (ii) universal gates realization (any 2) using pspice / any circuit simulation software.	
TOTAL HOURS		**60**

Suggested List of Students Activity

Proposed List of Student Activities for Digital Logic Design:

1. Research and present on the evolution of digital display technologies, including LED, LCD, OLED, and e-paper displays. Discuss their working principles, advantages, and applications in digital circuits.
2. Investigate and compare the specifications, availability, and costs of two different Analog-to-Digital Converter (ADC) and Digital-to-Analog Converter (DAC) ICs. Analyze their features and suitability for various applications.
3. Develop a block diagram approach for constructing a digital clock, frequency counter, or digital voltmeter using digital logic circuits. Estimate the cost of components required for the project and discuss potential challenges and solutions.
4. Explore the environmental impact of electronic waste (E-waste) and the proper disposal methods for printed circuit boards (PCBs) and integrated circuits (ICs). Conduct a seminar to raise awareness about E-waste management practices.
5. Design and simulate the functionality of a simple logic circuit using a modern software tool such as Logisim, Proteus, or LTspice.

Execution Guidelines:

- Limit each batch to a maximum of three students per activity.
- Assign activity No. 5 (logic circuit simulation) as mandatory for all batches.
- Assign any one activity from Nos. 1 to 5 or propose similar activities based on student interest and teacher discretion.

Reference

- Digital Electronics: Principles, Devices, and Applications" by Anil K. Maini

- “Digital Principles and Applications” Malvino& Leach, McGraw-Hill Education.
- "Modern Digital Electronics", R. P. Jain, McGraw Hill Education, 2009.
- ”Digital Principles and Design”, John F Wakerly, Donald D.Givone, Tata McGraw Hill, Edition.
- ”Digital Design Principles and Practices by Wakerly, John 5/e, Pearson Education 2018.
- "Fundamentals of Digital Circuits", A. Kumar, Prentice Hall India, 2016
- ” Fundamentals of logic design, Charles H. Roth, Jr. Cengage Learning, New Delhi.
- “Digital Design and Computer Architecture: ARM Edition, Sarah Harris and David Harris, 2015.

Web-based/Online Resources

- https://nptel.ac.in/
- https://www.electronicsforu.com/
- http://www.freebyte.com/electronics/
- https://www.edaplayground.com/
- https://www.circuitlogix.com
- http://www.vlab.co.in
- http://www.allaboutcircuits.com

Equipment / Facilities required to conduct the Practical Course

Software Requirement:

- Pspice software : https://www.pspice.com/
- LogiSim software : https://logisim.software.informer.com/2.7/
- Xcircuit Software : http://opencircuitdesign.com/xcircuit/
- Scilab : https://cloud.scilab.in/
- **Proprietor Software :**

Pspice, Multisim (available student version).

- Ltspice (available freely version).

Hardware Requirement:

- Digital Trainer Kit.
- Logic Gates: Basic logic gate ICs like 7400 (Quad 2-input NAND), 7402 (Quad 2-input NOR), and 7486 (Quad 2-input XOR).
- Flip-flops: Flip-flop ICs like 7474 (D flip-flop), 7476 (JK flip-flop), and 7473 (Dual JK flip-flop).

DRAFT

- Multiplexer and Encoder: IC74151 (8-to-1 multiplexer) and IC74148 (8-to-3 encoder).
- BCD to 7 segments Decoder: ICs like 7447 or 4511.
- 4-bit Ripple Counter: IC7493 or IC74161.
- 4-bit magnitude comparator 7485 IC: IC7485.
- H-Bridge motor driver IC (e.g., L293D).
- LED illumination control IC (e.g. LM317).
- Shift registers: IC7495, IC74194, or similar.
- For PLA: PLA device like 22V10.
- For PAL: PAL device like PAL16R8.
- For SRAM Operation: IC6264, DIP Switches for Address Selection / Data input/output.
- 1MHz,Oscilloscope
- ADC0804IC, DAC0808 IC
- Power Supply (0-30V).
- 5V , DC Motor
- Resistors, capacitors, diodes, LEDs.
- Breadboard, power supply, wires, and other necessary components for circuit construction and testing.

Continuous Assessment Pattern (CA1 & CA2)

Practical (50)

Allocation of Marks for CA1 & CA2 Practical Tests

Section	Description	Marks
1	Aim & Procedure	35
2	Execution and Result	15
TOTAL		**50**

DRAFT

BOARD PRACTICAL EXAMINATION

PART – A

1. Verify the truth tables of (i) Any 3 Logic Gates **(OR)** (ii) Any 3 Flip Flops.
2. Implement 2 input Boolean logic function using logic gates in
   (i) POS form **(OR)** (ii). SOP form.
3. Verify the truth tables of (i) Multiplexer IC74151 **(OR**) (ii) Encoder IC74148
4. Implement and verify the truth table using TTL/CMOS gates. ((i) BCD to 7 segments Decoder **(OR**) (ii) 4-bit Ripple Counter.
5. Verify the function of (i) 4-bit magnitude comparator 7485 IC
   (OR) (ii)Shift registers (ICs like 7495, 74194 etc.).

PART – B

6. Design a digital interface circuit to control the illumination of an LED and control the speed
   and direction of a DC motor using TTL/CMOS ICs.
7. Connect DAC0808 IC to convert digital to analog and ADC0804 IC to convert analog to digital. Show analog output via oscilloscope and digital via LEDs.
8. Implement the Boolean expression F = A'B'C' + A'BC + AB'C + ABC' using
   (i) PAL16R8 IC **(OR)** (ii) PLA22V10 IC.
9. Perform SRAM Operations through Reading, Writing using IC6264.
10. **Simulate** (i) Full Adder Circuits Using ICs **(OR)** (ii) universal gates realization (any 2) using pspice / any circuit simulation software.

SCHEME OF VALUATION

Section	Description	Marks
1	Aim (05), Procedure for the experiment from Part-A (30)	35
2	Aim (05), Procedure for the experiment from Part-B (30)	35
3	Execution of the experiment from Part-A **OR** Part-B	25
4	Viva voce	5
TOTAL MARKS		**100**

DRAFT

1052233440	C PROGRAMMING	L	T	P	C
Practicum		1	0	4	3

Introduction

In this course, students will learn the C programming language and its fundamental concepts. Also, they gain the knowledge to write simple C programs and undertake future courses that assume some background in computer programming. This course introduces programming principles using the C language. Students will learn C tokens, variables, data types, control structures, functions, arrays, pointers, structures and file concepts. Through hands-on students will develop proficiency in writing structured and efficient C programs to solve a variety of computational problems.

Course Objectives

The objectives of this course are enabling the students

- To learn problem solving skills.
- To gain knowledge of arrays and strings.
- To understand the concept of functions and their role in modular programming.
- To comprehend the basics of structures and its importance in application development.
- To recognize the importance of files and its related operations.

Course Outcomes

At the end of the course, students will be able

CO1: Demonstrate knowledge on C Programming concepts.

CO2: Develop simple programs in C using basic constructs.

CO3: Design modular C programs with reusable functions to improve code readability and maintainability.

CO4: Develop applications using structures and unions.

CO5: Apply programming skills to solve numerical problems and real-time problems.

Pre-requisites: Nil

DRAFT

CO/PO Mapping

CO / PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	3	3	3	1	1	1
CO2	3	3	3	3	1	2	2
CO3	3	3	3	3	1	2	1
CO4	3	3	3	3	1	2	2
CO5	3	3	3	3	1	1	2

Legend: 3-High Correlation, 2-Medium Correlation, 1-Low Correlation

Instructional Strategy

- Engage and Motivate: Instructors should actively engage students to boost their learning confidence.
- Real-World Relevance: Incorporate relatable, real-life examples and engineering applications to help students understand and appreciate course concepts.
- Interactive Learning: Utilize demonstrations and plan interactive student activities for an engaging learning experience.
- Application-Based Learning: Employ a theory-demonstrate-practice-activity strategy throughout the course to ensure outcome-driven learning and employability.
- Encourage Critical Analysis: Foster an environment where students can honestly assess experiment outcomes and analyze potential sources of error in case of discrepancies.

DRAFT

Assessment Methodology

	Continuous Assessment (40 marks)					End Semester Examination (60 marks)
	CA1		CA2		CA3	
Mode	Written Test First 50% of Theory	Practical Test (50% of Exercise)	Written Test (Next 50% of Theory)	Practical Test (50% of Exercise)	Model Practical Exam	Practical Exam
Duration	2 hours	2 hours	2 hours	2 hours	3 hours	3 hours
Exam Marks	50	50	50	50	100	100
Converted to	20		20		10	60
Practical log book / Observation Note book with bonafide	10					
Final Marks	40					60

Note:

DRAFT

- Continuous Assessment tests (CA1 and CA2) should be conducted for both Theory and Practical portions. **Best of one** will be considered for 20 marks for internal assessment.
- Continuous Assessment Model practical examination (CA3) should be conducted as per the end semester question pattern after completion of all the portions. The marks awarded shall be converted to 10 Marks.
- Practical logbook / Observation notebook with bonafide carries 10 Marks.

1052233440	C PROGRAMMING	L	T	P	C
Practicum		1	0	4	3

Unit I	INTRODUCTION TO C	
Overview of C: Basic Structure of C program, Compiling and Executing a C program. C Basics: Constants, Variable, Operators and Datatypes -Character Set - C Tokens -Keywords and Identifiers – Declaration and Use of Variables – Managing Input and Output Operations: Formatted and Unformatted Input and Output statements.		3
Ex No 1: Write a C program to calculate the simple interest and compound interest. **Ex No 2**: Write a C program to find the area of a circle and a rectangle (use preprocessing directory for defining pi value).		12
Unit II	**CONTROLSTRUCTURES AND LOOPING**	
Decision Making and Branching: Simple if Statement - if-else Statement – Nested if-else Statements - else if ladder – Switch case statement. Looping: While Statement – do..while statement – for loop statement-break and continue statement- goto statement.		3
Ex No 3: Write a C program to find the largest of three numbers. **Ex No 4:** Write a C program to generate all prime numbers from 1 to N.		12
Unit III	**ARRAYS AND FUNCTIONS**	
Arrays: One-dimensional Arrays- Declaration of One-dimensional Arrays - Initialization of One-dimensional Arrays-Two-dimensional Arrays- Declaration of Two-dimensional Arrays - Initialization and accessing Two-dimensional Arrays.		3

DRAFT

User-Defined Functions: Need for Functions -Elements of User Defined Functions –Functions Types – Call by Value-Call by Reference-Passing Arrays to Functions-Recursion **Ex No 5**: Write a C program to demonstrate matrix addition and transpose operations. **Ex No 6:** Write a C program to find factorial of a given number using recursion.		12
UNIT IV	**STRINGSANDPOINTERS**	
Strings: Declaring and Initializing String Variables –String Built-in Functions. Pointer: Declaring, Initialization and Accessing a pointer Variable-Pointer Expressions – Pointer Arithmetic – Pointer to Pointer - Pointer to an Array.		3
Ex No 7: Write a C program to find the length and reverse a string using pointers. **Ex No 8**: Write a C program for implementing linear search using pointer to an array.		12
UNIT V	**STRUCTURESANDFILEMANAGEMENT**	
Structures and Union: Introduction -Defining a Structure -Declaring Structure Variables – Accessing Structure Members – structure initialization -Array of Structures-Union. File Management in C: Introduction – File Types- Defining File Modes -Opening and Closing a File – File Operations.		3
Ex No 09: Write a C program to collect and print students details like name, marks, etc. and then calculate total and average mark using structure. **Ex No 10:** Write a C program to count the number of characters, words and lines in a file.		12
TOTAL HOURS		**75**

DRAFT

TEXT BOOKS

1. Reema Thareja, “Programming in C”, Oxford University Press, Second Edition, 2015.
2. Pradip Dey, Manas Ghosh, “Fundamentals of Computing and Programming in C”, First Edition, Oxford University Press, 2009.
3. Anita Goel and Ajay Mittal, “Computer Fundamentals and Programming in C”, Pearson Education in South Asia, 2016.

4. E.Balaguruswamy, “Programming in ANSI C”, Tata McGraw-Hill, Third Edition Edition,2012.

REFERENCE BOOKS

1. Paul Deitel and Harvey Deitel, “C How to Program”, Eighth Edition, Pearson Publication,2016.
2. Juneja, B. L and Anita Seth, “Programming in C”, Cengage Learning India Pvt. Ltd., 2011.

WEB-BASED/ONLINE RESOURCES

- https://www.w3schools.com/c/
- https://www.programiz.com/c-programming
- https://www.tutorialspoint.com/cprogramming/index.htm
- https://egyankosh.ac.in/

Suggested List of Students Activity

- Presentation/Seminars by students on any recent technological developments based on the course Programming assignments
- Periodic class/online quizzes conducted based on the course.
- Blended learning activities to explore the recent trends and developments in the field.

Equipment / Facilities required to conduct the Practical Portion

1. **Hardware(s) Requirement:**
   - Desktop / Laptop
   - Printer
2. **Software(s) Requirement:**
   - Windows / Linux Operating System
   - Code Blocks / Turbo C

Continuous Assessment Pattern (CA1 & CA2)

Theory (50)

Allocation of Marks for CA1 & CA2 Theory Tests

Section	Description	Marks

DRAFT

Part – A	11 questions to be answered out of 12	11 x 2 marks	22
Part – B	4 questions to be answered out of 6	4 x 7 marks	28
TOTAL			**50**

Practical (50)

Allocation of Marks for CA1 & CA2 Practical Tests

Section	Description	Marks
1	Aim & Program	35
2	Execution and Result	15
TOTAL		**50**

DRAFT

BOARD PRACTICAL EXAMINATION

PART – A

1. Write a C program to calculate the simple and compound interest.
2. Write a C program to find the area of a circle and a rectangle (use preprocessing directory for defining pi value).
3. Write a C program to find the largest of three numbers.
4. Write a C program to generate all prime numbers from 1 to N..
5. Write a C program to find factorial of a given number using recursion.

PART - B

6. Write a C program to demonstrate matrix addition and transpose operations.
7. Write a C program to find the length of a string and reverse a string using pointers
8. Write a C program for implementing linear search using pointer to an array.
9. Write a C program to collect and print students details like name, marks, etc. and then calculate total and average mark using structure.
10. Write a C program to count the number of characters, words and lines in a file.

SCHEME OF VALUATION		
SNO	**ALLOCATION**	**MARKS**
1	Aim (05) ,Program from Part – A (30)	35
2	Aim (05) ,Program from Part – B (30)	35
3	Executing any one program (Part A or Part –B)	15
4	Output	10
5	Viva Voce	05
6	Total	100

DRAFT

1052233540	WEB DESIGNING	L	T	P	C
Practicum		1	0	4	3

Introduction:

This course provides an introductory overview of the principles and practices of web design. Students will learn the fundamentals of creating visually appealing and user-friendly websites. Through a combination of theory and hands-on projects, students will explore various aspects of web design through basic coding languages such as HTML, CSS and client-side scripting language like JavaScript.

Course Objectives:

1. Learn the syntax, structure, and basic elements of HTML, including tags, attributes, and semantic markup.
2. Gain proficiency in creating and organizing content using HTML elements such as headings, paragraphs, lists, and links and media elements.
3. Explore CSS syntax, selectors, properties, and values for styling HTML elements.
4. Learn how to apply CSS styling to text, backgrounds, borders, and other page elements.
5. Understand the role and importance of client-side scripting in web development.
6. Gain proficiency in JavaScript syntax, data types, and operators.
7. Learn how to manipulate the Document Object Model (DOM) using JavaScript.
8. Explore various techniques for handling user interactions and events on web pages.
9. Learn how to validate form inputs and handle form submissions using JavaScript.

Course Outcomes:

On successful completion of this course, the student will be able to

CO1: Develop well-structured web pages using HTML.

CO2: Develop web pages using HTML links and media elements.

CO3: Apply CSS to effectively style and format HTML elements to create visually appealing web pages.

CO4: Write and deploy Javascript code to solve practical web design problems

CO5: Utilize JavaScript to add interactivity and dynamic behavior to web pages including event handling, DOM manipulation, and form validation.

Pre-requisites:

Knowledge of web browsers, websites, and basic internet terminologies.

CO/PO Mapping

CO / PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2	3	3	2	1	1	2
CO2	2	3	3	3	1	1	2
CO3	2	3	3	2	1	1	2
CO4	2	3	3	3	1	1	2
CO5	2	3	3	3	1	1	2

Legend: 3-High Correlation, 2-Medium Correlation, 1-Low Correlation

Instructional Strategy:

- It is advised to assign practical projects that allow students to apply concepts learned in class. Projects could include designing and coding websites from scratch, creating prototypes, or redesigning existing websites to improve usability and aesthetics.
- Conduct of interactive demos to help students learn specific techniques and tools. Provide step-by-step guidance and encourage questions and participation.
- Real-world examples of websites to understand design principles, usability issues, and best practices and discuss how different design choices impact user experience and business goals.

Assessment Methodology:

	Continuous Assessment (40 marks)					End Semester Examination (60 marks)
	CA1		CA2		CA3	
Mode	Written Test (First 50% of Theory)	Practical Test (First 50% of Exercise)	Written Test (Next 50% of Theory)	Practical Test (Next 50% of Exercise)	Model Practical Exam	Practical Exam
Duration	2 hours	2 hours	2 hours	2 hours	3 hours	3 hours
Exam Marks	50	50	50	50	100	100
Converted to	20		20		10	60
Practical log book / Observation Note book with bonafide	10					
Final Marks	40					60

Note:

- CA1 and CA2 Assessment tests should be conducted for both Theory and Practical portions. **Best of one** will be considered for 20 Marks for internal assessment.
- CA3 Model Practical examination should be conducted as per the end semester question pattern after completion of all the portions. The marks awarded shall be converted to 10 Marks.
- Practical Logbook / Observation Notebook with Bonafide Carries 10 Marks.

DRAFT

1052233540	WEB DESIGNING	L	T	P	C
Practicum		1	0	4	3

Unit I HTML & HTML Tags	
Introduction to HTML: HTML editors-HTML5- HTML Elements & Attributes. HTML Tags -Page Formatting Tags: DOCTYPE Tag- html tag- head, title, body, meta, script, style tags Text Formatting Tags: Heading Tags- Paragraph Tags- Horizontal rules- Line breaks- Superscript- Subscript- Underline- Italic- Bold- Emphasis- del tags HTML List Tags: Unordered List- Ordered List-Definition List HTML Link Tags & attributes- HTML Table Tags: table, th, tr, td, colspan, rowspan	**3**
1. Write a HTML code to display welcome text using different text formatting tags.(Use h1-h6, bold, italic, underline, strikethrough,div,p,pre tags) **(HTML Basic tags)**	**6**
2. Design a HTML page to list the computer languages where each language is a link. Prepare separate HTML documents for each language and call them in the appropriate link. **(Lists and Links)**	**6**
Unit II HTML & CSS	
HTML Document Layout Tags: header, footer, main, section tags HTML Media Tags- Images- <img> tags & attributes, Image Mapping using <map> and <area> tags - background images- Other media tags- <audio>, <video>, <iframe>, <embed>, <svg>, <canvas> CSS- Introduction- Need for CSS- Syntax- Selector- Declaration, Property, value- Types of CSS selector: Class, id, pseudo-class, attribute, universal selector- Types of style sheets: Internal - External- Inline- Color values. CSS background properties- Border properties- margin properties- padding	**4**

DRAFT

properties- height, width properties- CSS Text properties- CSS Fonts properties	
3. Write a HTML program to display the image of a computer as a link to the web page describing the components of computers. **(Images and link tags)**	**6**
4. Develop a web page using CSS to create a timetable for the class using different border style. **(Table tags and internal style sheets)**	**6**
5. Design a webpage of your college with attractive background color, text-color, font-face, an image by using external CSS formatting .**(External Stylesheets)**	**6**
Unit III JavaScript	
Scripting Languages: Client Side Scripting- Server Side Scripting- Need for javascript - structure of javascript - Variables- Datatypes- String- Number- Boolean- Undefined- Null JavaScript Objects: Array- String- Date- Math- Number- Boolean- User Defined Objects. Operators: Arithmetic - Assignment - Comparison - String - Logical - Bitwise - Ternary - Type Conditional Statements: if, if-else, else-if, switch. Loop statements- for, while, do-while- break- continue statements. JavaScript functions: definition- parameters-function call- function invocation	**4**
6. Write a JavaScript program to create a clock in 24 hours format using Date Object. (Do not include AM/PM) (**JavaScript Objects and Functions**)	**6**
7. Write a JavaScript program to control (play, pause, stop) the audio/video in a web page. **(JavaScript User defined Objects and Media Tags)**	**6**
Unit IV JavaScript Advanced	
Forms: Form tag- action, method, target, auto complete attributes, input tag, type attribute values- text, radio, checkbox, button, submit, password, other attributes for input tag -id, name, value , size, required. Special tags in forms -textarea tag, select tag, button tag, label tag. Message Boxes: Dialog Box- Alert Box- Confirm Box- Prompt Box JavaScript Document Object Model: Methods of Document object- Javascript Events- Event Handlers- Mouse events- Keyboard Events- Form Events- Window	**4**

DRAFT

Events JavaScript Form Validation- Email validation	
8. Write a JavaScript program to change the color of a web page to the color typed by the user in the text box. **(DOM)**	**6**
9.Write a JavaScript program to develop a simple calculator (with basic arithmetic operations like add, subtract, multiply, divide, equal to) by getting two` numbers in two text boxes, buttons for operations and display the result in the third text box. **(Event Handling)**	**6**
10. Create a form with text fields such as username, mail id, password, retype password, gender (radio), languages known (check box), Describe yourself (textarea), submit button and perform form validation such as username must not be less than 8 characters, mail id should contain @ symbol, password and retype password must be the same. **(Forms & Form validation)**	**6**
Total Periods	**75**

Text and Reference Books:

1. "Douglas E. Comer" "The Internet Book", Prentice Hall.
2. "Terry Felke-Morris" "Web Development and Design Foundations with HTML5", Pearson.
3. "Thomas A. Powell, Fritz Schneider" "HTML & CSS: The Complete Reference", Tata McGrawHill.
4. "Thomas Powell, Fritz Schneider" "Java Script: The Complete Reference", Tata McGraw-Hill.
5. "Jennifer Niederst Robbins" "Learning Web Design: A Beginner's Guide to HTML, CSS, JavaScript and Web Graphics" , O'Reily.
6. "Paul McFedries" "Web Design Playground: HTML & CSS the Interactive Way", Manning Publications.

Web-based/Online Resources:

https://www.w3schools.com/

https://www.freecodecamp.org/

https://javascript.info/

https://www.coursera.org/

https://www.codecademy.com/

Learn JavaScript Tutorial - javatpoint

DRAFT

Suggested List of Students Activity:

- Quizzes/ Seminars/ Presentations to students to evaluate their learning concepts.
- Mini Project based learning to work on a website project incorporating HTML, CSS, and JavaScript as an extension to real life applications.
- Conduct code reviews to provide feedback on HTML, CSS, and JavaScript code and to debug and troubleshoot it.

Equipment / Facilities required to conduct the Practical Course

Hardware Required:

1. Desktop Computers / Laptop
2. Laser Printer

Software Required:

1. Notepad / Notepad++ / Dreamweaver
2. Any Browser.

Continuous Assessment Pattern (CA1 & CA2)

Theory (50)

Allocation of Marks for CA1 & CA2 Theory Tests

Section	Description	Marks	
Part – A	11 questions to be answered out of 12	11 x 2 marks	22
Part – B	4 questions to be answered out of 6	4 x 7 marks	28
TOTAL			**50**

Practical (50)

Allocation of Marks for CA1 & CA2 Practical Tests

Section	Description	Marks
1	Aim & Program	35
2	Execution and Result	15
TOTAL		**50**

DRAFT

BOARD PRACTICAL EXAMINATION

PART – A

1. Write a HTML code to display welcome text using different text formatting tags.(Use h1-h6, bold, italic, underline, strikethrough,div,p,pre tags) **(HTML Basic tags)**

2. Design a HTML page to list the computer languages where each language is a link. Prepare separate HTML documents for each language and call them in the appropriate link. **(Lists and Links)**

3. Write a HTML program to display the image of a computer as a link to the web page describing the components of computers. **(Images and link tags)**

4. Develop a web page using CSS to create a timetable for the class using different border style. **(Table tags and internal style sheets)**

5. Design a webpage of your college with attractive background color, text-color, font-face, an image by using external CSS formatting .**(External Style sheets)**

PART – B

6.Write a JavaScript program to create a clock in 24 hours format using Date Object. (Do not include AM/PM) (**JavaScript Objects and Functions**)

7. Write a JavaScript program to control (play, pause, stop) the audio/video in a web page. **(JavaScript User defined Objects and Media Tags)**

8. Write a JavaScript program to change the color of a web page to the color typed by the user in the text box. **(DOM)**

9. Write a JavaScript program to develop a simple calculator (with basic arithmetic operations like add, subtract, multiply, divide, equal to) by getting two numbers in two text boxes, buttons for operations and display the result in the third text box. **(Event Handling)**

10.Create a form with text fields such as username, mail id, password, retype password, gender (radio), languages known (check box), Describe yourself (textarea), submit button and perform form validation such as username must not be less than 8 characters, mail id should contain @ symbol, password and retype password must be the same. **(Forms & Form validation)**

SCHEME OF VALUATION		
SNO	**ALLOCATION**	**MARKS**

DRAFT

1	Aim (05) , Program from Part – A (30)	35
2	Aim (05) , Program from Part – B (30)	35
3	Executing any one program (Part A or Part –B)	15
4	Output	10
5	Viva Voce	05
6	Total	100

1052233640	OPERATING SYSTEMS	L	T	P	C
Practicum		1	0	2	2

Introduction

Students have to be conversant with computer, its terminology and functioning. The heart of a computer is based around its Operating System. An operating system acts as an interface between the user of a computer and the computer hardware. The processor deals with request coming from all directions asynchronously. The operating system has to deal with the problems of contention, resource management and both program and user data management, and provide a useful no-wait user interface. The course provides clear vision, understanding and working of Operating Systems.

Course Objectives

On completion of the following units of syllabus contents, the students must be able to

- To understand the purpose, goals, functions and evolution of Operating Systems. Login and logoff Procedures
- To know how to use of General purpose and communication commands
- To study the use of Search patterns, simple filters and advanced filters
- To know the details of process status
- To understand shell scripts, define the elements of the shell script and write shell script for various problems.

Course Outcomes

After successful completion of this course, the students should be able to

CO1: understand functional architecture of an operating system.

CO2: distinguish CPU scheduling algorithms.

CO3: analyze process coordination.

CO4: classify File System and directory implementations.

CO5: Resource Management

Pre-requisites: Nil

CO/PO Mapping

CO / PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2	2	2	2	2	---	---
CO2	2	3	3	2	2	---	---
CO3	2	2	2	3	3	---	---
CO4	2	2	3	2	2	---	---
CO5	2	3	2	3	3	---	---

Legend: 3-High Correlation, 2-Medium Correlation, 1-Low Correlation

Instructional Strategy

- Engage and Motivate: Instructors should actively engage students to boost their learning confidence.
- Real-World Relevance: Incorporate relatable, real-life examples and engineering applications to help students understand and appreciate course concepts.
- Interactive Learning: Utilize demonstrations and plan interactive student activities for an engaging learning experience.
- Application-Based Learning: Employ a theory-demonstrate-practice-activity strategy throughout the course to ensure outcome-driven learning and employability.

DRAFT

- Encourage Critical Analysis: Foster an environment where students can honestly assess experiment outcomes and analyze potential sources of error in case of discrepancies.

Assessment Methodology

	Continuous Assessment (40 marks)					End Semester Examination (60 Marks)
	CA1		CA2		CA3	
Mode	Written Test (First 50% of Theory)	Practical Test (50% of Exercise)	Written Test (Next 50% of Theory)	Practical Test (50% of Exercise)	Model Practical Exam	Practical Exam
Duration	2 hours	2 hours	2 hours	2 hours	3 hours	3 hours
Exam Marks	50	50	50	50	100	100
Converted to	20		20		10	60
Practical log book / Observation Note book with bonafide	10					
Final Marks	40					60

Note:

- Continuous Assessment tests (CA1 and CA2) should be conducted for both Theory and Practical portions. **Best of one** will be considered for 20 marks for internal assessment.
- Continuous Assessment Model practical examination (CA3) should be conducted as per the end semester question pattern after completion of all the portions. The marks awarded shall be converted to 10 Marks.
- Practical logbook / Observation notebook with Bonafide carries 10 Marks.

DRAFT

1052233640	OPERATING SYSTEMS	L	T	P	C
Practicum		1	0	2	2

PART A	INTRODUCTION TO OS	
	Introduction to operating system: Basics of Operating system- types of operating system- operating system services – operating system structures – Process Management – Process scheduling Basics of Linux OS: Entering and Exiting from a Linux System – User Accounts – Different shells – Learn the syntax and usage of Directory Management Commands – Check the process status – process management commands – search patterns Deadlocks – Virtual Memory – Page Replacement Algorithms	8
	Ex No:1 Write a syntax and execute the directory management commands : ls, cd, pwd, mkdir, rmdir **Ex No:2** Write a syntax and execute the file management commands such as cat, chmod, cp, mv, rm, more **Ex No:3** Write a syntax and execute the general purpose commands : wc, cal,	15

date, who, tty, ln **Ex No:4** Using the simple filters verify pr, head, tail, cut, paste, nl, sort grep, egrep, fgrep, write and wall		
PART B	**SHELL SCRIPTS**	
File operations (New, Open, Close, Save, Save and Exit, Print) – Text Editing operations (inserting ,deleting ,finding, replacing, copying and moving)- use of shell scripts – Numerical operations – Looping – Swapping techniques – string operations- using command line arguments – filters-date function- Relational Operations -Logical Operations – Boolean operations – Basic Arithmetic operations – case statement – search directory or file .		7
Ex No 5: Write a shell script that accepts a numerical value N and find sum . **Ex No 6:** Write a shell script to find factorial of the given number . **Ex No 7:** Write a shell script to perform arithmetic calculator using case statement. **Ex No 8:** Write a shell script using command line arguments and reports on whether it is a directory, file or something else.		15
TOTAL PERIODS		**45**

References

1. TEXT BOOKS:

1. Abraham Silberchatz, Peter B. Galvin, Greg Gagne, Operating System Principles, 9th Edition, John Wiley &Sons,2018.

2.References

1. William Stallings, Operating Systems – Internal and Design Principle", 9thEdition, Pearson Education/PHI,2018.
2. D.M. Dhamdhere, Operating systems - A Concept based Approach, 3rd Edition, TMH,2017.
3. Charles Crowley, Operating Systems - A Design Approach, 1st Edition, TMH,2017.
4. Andrew S Tanenbaum, Modern Operating Systems, 3rd Edition, Pearson/PHI,2014.

3. WEB REFERENCES:

1. https://nptel.ac.in/courses/106/105/106105214/
2. https://ocw.mit.edu/courses/6-828-operating-system-engineering-fall-

DRAFT

2012/pages/lecture-notes-and-readings/

3. https://www.geeksforgeeks.org/what-is-an-operating-system/
4. https://www.w3schools.in/operating-system/intro

Suggested List of Students Activity

- Presentation/Seminars by students on any recent technological developments based on the course.
- Periodic class/online quizzes conducted based on the course.
- Blended learning activities to explore the recent trends and developments in the field.

Equipment / Facilities required to conduct the Practical Portion

1. **Hardware Requirement:**
   - Desktop Computers
   - Printer
2. **Software Requirement:**
   - Linux Operating System

DRAFT

Continuous Assessment Pattern (CA1 & CA2)

Theory (50)

Allocation of Marks for CA1 & CA2 Theory Tests

Section	Description	Marks	
Part – A	11 questions to be answered out of 12	11 x 2 marks	22
Part – B	4 questions to be answered out of 6	4 x 7 marks	28
TOTAL			**50**

Practical (50)

Allocation of Marks for CA1 & CA2 Practical Tests

Section	Description	Marks
1	Aim & Program	35
2	Execution and Result	15
TOTAL		**50**

BOARD PRACTICAL EXAMINATION

PART – A

Ex No:1 Write a syntax and execute the directory management commands : ls,cd,pwd,mkdir,rmdir

Ex No:2 Write a syntax and execute the file management commands such as cat, chmod, cp, mv ,rm,

more

Ex No:3 Write a syntax and execute the general purpose commands : wc,cal,date,who,tty,ln

Ex No:4 Using the simple filters verify pr,head,tail,cut,paste,nl,sort grep,egrep,fgrep,write and wall

PART – B

Ex No 5: Write a shell script that accepts a numerical value N and find sum .

Ex No 6: Write a shell script to find factorial of the given number .

Ex No 7: Write a shell script to perform arithmetic calculator using case.

Ex No 8: Write a shell script using command line arguments and reports on whether it is a directory,

file or something else.

SCHEME OF VALUATION		
SNO	**ALLOCATION**	**MARKS**
1	Aim (05) ,Program from Part – A (30)	35
2	Aim (05) ,Program from Part – B (30)	35
3	Executing any one program (Part A or Part –B)	15
4	Output	10
5	Viva Voce	05

DRAFT

6	Total	100

DRAFT

1052234110	Computer Networks and Security	L	T	P	C
Theory		3	0	0	3

Introduction

The course aims to groom the students to gain concepts, knowledge and skillsrequired to work on Computer Networking and Security industry. Course curriculum hasbeen designed to give overview and use cases of Data Communication, Layered Networks,Inter-networking technology/protocols and Computer Security is covered and this will help toprepare the students to keep pace with computer networking and security industry trends.

Course Objectives

The objective of this course is to enable the student to

- Understand the concept of data communication
- To know the functions and protocols of each layer of OSI and TCP/IP protocol suite.
- To visualize the end-to-end flow of information.
- Understand the main principles of computer and network security.
- Know different networking devices and their practical usages.
- Know the IP addressing and its mechanisms.
- Identify the attacks and threats.
- Study about Cryptography and different Cryptography Algorithms.

Course Outcomes

After successful completion of this course, the students should be able to

CO1: Remember the fundamentals of Computer Networks.

CO2: Identify core networking and infrastructure components, and the services

CO3: Examine the different networking applications

CO4: Understand fundamental properties of computer security, such as Authentication, Authorization, and Data confidentiality and Integrity.

CO5: Identify the concept of the internet and security issues.

Pre-requisites: Nil

DRAFT

CO/PO Mapping

CO / PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	3	3	1	1	1	1
CO2	3	3	3	1	1	1	2
CO3	3	3	3	1	1	1	1
CO4	3	3	3	1	1	1	2
CO5	3	3	3	1	1	1	2

Legend: 3-High Correlation, 2-Medium Correlation, 1-Low Correlation

Instructional Strategy

- Engage and Motivate: Instructors should actively engage students to boost their learning confidence.
- Real-World Relevance: Incorporate relatable, real-life examples and engineering applications to help students understand and appreciate course concepts.
- Interactive Learning: Utilize demonstrations and plan interactive student activities for an engaging learning experience.
- Application-Based Learning: Employ a theory-demonstrate-practice-activity strategy throughout the course to ensure outcome-driven learning and employability.
- Simulation and Real-World Practice: Conduct demonstrations and hands-on activities in a simulated environment, transitioning to real-world scenarios when possible.
- Encourage Critical Analysis: Foster an environment where students can honestly assess experiment outcomes and analyze potential sources of error in case of discrepancies.

DRAFT

Assessment Methodology:

	Continuous Assessment (40 marks)			End Semester Examination (60 marks)
	CA1	CA2	CA3	
Mode	Written Test (Unit I & II)	Written Test (Unit III & IV)	Model Theory Exam	Theory Exam
Duration	2 Hours	2 Hours	3 Hours	3 Hours
Exam Marks	50	50	100	100
Converted to	20	20	20	60
Final Marks	40			60

Note:

- CA1 and CA2 Assessment tests should be conducted.CA1 covers unit 1 and unit 2, CA2 covers unit 3 and unit 4. Best of one will be considered for 20 Marks for internal assessment.
- CA3 Model examination should be conducted as per the end semester question pattern after completion of all the portions. The marks awarded shall be converted to 20 Marks.

DRAFT

1052234110	Computer Networks and Security	L	T	P	C
Theory		3	0	0	3

Unit I	DATACOMMUNICATIONS	9

Data Communication: Components of a data communication– Data flow: Simplex - Half duplex - Full duplex; Networks- Network Criteria – Types of Connections: Point to Point – Multipoint; Topologies: Star, Bus, Ring, Mesh, Hybrid – Advantages and Disadvantages of each topology.

Types of Networks: Need of Computer Networks – LAN – MAN – WAN – Internet – Intranet – Internet Service Providers (ISP) – Client-server – Peer to Peer – Wi-Fi – Bluetooth – Protocols and Standards

Network Devices: Switches – Bridges – Routers – Gateways;

Unit II	DATALINK LAYER	9

Network Models: OSI Model – Layered Architecture – Function of Layers – TCP/IP Protocol Suite

802.X Protocols: Concepts and PDU format of CSMA/CD(802.3) – Token ring (802.5) – Ethernet – Types of Ethernet(Fast Ethernet, gigabit Ethernet, High speed Ethernet 10GE to 800GE) – Comparison between 802.3 and 802.5

IP Addressing: Dotted Decimal Notation – Subnetting and Supernetting – Ipv4 – Ipv6

Unit III	NETWORK,TRANSPORT AND APPLICATION LAYER	9

Network Layer Protocol: IP – Interior Gateway Protocols: IGMP, ICMP, ARP, RARP

Overview of TCP/IP - Transport Layer Protocols: Stop and wait protocol-Go Back N Protocol – Connection Oriented and Connectionless Service – Sockets – TCP and UDP

Application Layer Protocols: FTP – HTTP – SMTP – POP -DNS - Telnet

Unit IV	NETWORK SECURITY AND IP SECURITY	10

Network Security: Definition – Need of Network Security – Principles of Security – Attacks – Types of Attacks – Criminal Attacks – Legal Attacks – Passive and Active Attacks – Software Supply Chain Attacks

Cryptography: Definition – Symmetric Encryption Principles – Symmetric Block Encryption algorithms – DES - Digest Function – Public key cryptography principles – RSA– Digital Signature

Internet Security: Email Security: PGP – S/MIME; IP Security: Overview – Architecture

Unit V	NETWORK SECURITY APPLICATIONS	8

DRAFT

Hackers Techniques: Historical hacking techniques and Open sharing – Bad Passwords – Advanced techniques – Viruses – Worms – Trojan Horses – SPAM **Security Mechanisms:** Introduction – Types of Firewalls – Packet Filters – Application Gateways – Limitation of Firewalls **Intrusion:** Intruders – Intruder detection – Classification of Intruder - Detection Systems – Honeypots	
TOTAL PERIODS	**45**

Text Books and References

1. Behrouz A. Forouzan, “Data communication and Networking”, Tata McGraw-Hill, 2004.

2. Andrew S. Tanenbaum “Computer Networks”, Prentice Hall Edition, 5th Edition.

3. William Stallings, “Data and Computer Communications”, Eighth Edition, Pearson Education, 2011.

4. James F. Kurose, Keith W. Ross, “Computer Networking, A Top-Down Approach Featuring the Internet”, Third Edition, Pearson Education, 2006.

5. Larry L. Peterson, Bruce S. Davie, “Computer Networks: A Systems Approach”, Fifth Edition, Morgan Kaufmann Publishers Inc., 2011.

Web-based/Online Resources

Network Devices
https://www.pynetlabs.com/network-devices-and-its-various-types/

Socket Connections
https://learn.microsoft.com/en-us/dotnet/fundamentals/networking/sockets/socket-services

Hacking Techniques
https://portswigger.net/research/top-10-web-hacking-techniques-of-2021

Suggested List of Students Activity

- Presentation/Seminars by students on any recent technological developments based on the course.
- Periodic class Assessments conducted based on the course.
- Blended learning activities to explore the recent trends and developments in the field.

Continuous Assessment Pattern (CA1 & CA2)

DRAFT

Theory (50)

Allocation of Marks for CA1 & CA2 Theory Tests

Section	Description	Marks	
Part – A	11 questions to be answered out of 12	11 x 2 marks	22
Part – B	4 questions to be answered out of 6	4 x 7 marks	28
TOTAL			**50**

1052234230	**Data Structures Using Python**	L	T	P	C
Practicum		3	0	2	4

Rationale

Data structure is a subject of primary importance in Information and Communication Technology. Knowledge of data structures is essential for implementation of efficient algorithms and program development. Learning data structures with Python offer flexibility and ease of programming with many built in data structures and libraries.

Course Objectives

The objective of this course is to

1. Provide the knowledge of various types of data structures
2. Familiarize with the representation of data structures
3. Use various data structures in organizing data
4. Reinforce theoretical concepts by writing relevant programs
5. Gain knowledge in practical applications of data structures

Course Outcomes

After successful completion of this course, the students should be able to

CO1: Understand the fundamental data structures

CO2: Understand the concepts of linked lists

CO3: Apply the operations of stack and queue

CO4: Illustrate tree structure and apply trees traversal techniques

DRAFT

CO5: Implement various sorting and searching techniques

Pre-requisites

Knowledge in C and python programming.

CO/PO Mapping

CO / PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	3	3	3	-	-	-
CO2	3	2	2	1	-	-	-
CO3	3	3	3	2	-	-	-
CO4	3	3	3	2	-	-	-
CO5	3	3	2	2	-	-	-

Legend: 3-High Correlation, 2-Medium Correlation, 1-Low Correlation

Instructional Strategy

- Engage and Motivate: Instructors should actively engage students to boost their learning confidence.
- Real-World Relevance: Incorporate relatable, real-life examples and engineering applications to help students understand and appreciate course concepts.
- Interactive Learning: Utilize demonstrations and plan interactive student activities for an engaging learning experience.
- Application-Based Learning: Employ a theory-demonstrate-practice-activity strategy throughout the course to ensure outcome-driven learning and employability.
- Encourage Critical Analysis: Foster an environment where students can honestly assess experiment outcomes and analyze potential sources of error in case of discrepancies.

DRAFT

Assessment Methodology

	Continuous Assessment (40 marks)					End Semester Examination (60 marks)
	CA1		CA2		CA3	
Mode	Written Test (First 50% of Theory)	Practical Test (First 50% of Exercise)	Written Test (Next 50% of Theory)	Practical Test (Next 50% of Exercise)	Model Theory Exam	Theory Exam
Duration	2 hours	2 hours	2 hours	2 hours	3 hours	3 hours
Exam Marks	50	50	50	50	100	100
Converted to	20		20		10	60
Practical log book / Observation Note book with bonafide	10					
Final Marks	40					60

Note:

DRAFT

- Continuous Assessment tests (CA1 and CA2) should be conducted for both Theory and Practical portions. **Best of one** will be considered for 20 marks for internal assessment.
- Continuous Assessment Model Theory examination (CA3) should be conducted as per the end semester question pattern after completion of all the portions. The marks awarded shall be converted to 10 Marks.
- Practical log book / Observation note book with bonafide carries 10 Marks.

DRAFT

1052234230	Data Structures Using Python	L	T	P	C
Practicum		3	0	2	4

UNIT I	INTRODUCTION TO DATA STRUCTURES	
Data structures – Introduction, classification of data structures : primitive and non-primitive data structures with python examples – linear and nonlinear data structures with python examples. Operations on data structures. **Abstract data types** - Introduction, abstractions, Abstract data types, example of abstract data type (student, date), Defining the ADT, Using the ADT, Implementing the ADT. **Algorithm Analysis** – space complexity, time complexity, Asymptomatic notations: Big-O notation.		9
Ex.No	Name of the Experiment	
1	Write a program to implement any one python data structure with the following operations A) Create B) Add elements C) Access elements D) Remove elements	2
UNIT II	**LINEAR DATA STRUCTURES - LINKED LISTS**	
Linked List - Terminology: node, address, information, null pointer, empty list, Types – singly linked lists: creating nodes, traversing the nodes, searching for a		8

node, prepending nodes, removing nodes - doubly linked list & circular linked list – organization - operations: traversal, searching, adding nodes, removing nodes (concepts only, no implementations)		
Ex.No	Name of the Experiment	
2	Write a python program to implement a singly linked list a) create a singly linked list b) add element to singly linked list c) Remove element from singly linked list	4
UNIT III	**LINEAR DATA STRUCTURES – STACK & QUEUE**	
Stacks- Overview of stack, Implementation of stack using python list: push, pop, display. Stack applications: balanced delimiters, evaluating postfix expressions. Recursion - Properties of recursion - Recursive functions: Factorials, Recursive call tree. **Queues-** Overview of queue **-** Implementing the queue and its operations using python list - Applications of queues - Circular queue and Priority queue (concepts only)		10
Ex.No	Name of the Experiment	
3	Write a python program to implement stack	8
4	Write a python program to implement queue	
UNIT IV	**NON-LINEAR DATA STRUCTURES - TREES**	
Tree - Terminology: node, edge, parent, children, path, level of a node, depth of a node, height of a tree – Binary tree: full binary tree, complete binary tree – Linear representation of binary tree - binary tree traversals: in-order, pre-order, post-order. Binary Search Tree – Introduction, Creation of a Binary Search tree without duplicate node, Applications.		9
Ex.No	Name of the Experiment	
5	Write the python program for pre-order traversal of a binary tree	4
UNIT V	**SEQUENTIAL STORAGE REPRESENTATION – SORTING & SEARCHING**	
Sorting- Introduction to different sorting techniques - Bubble sort, Insertion sort,		9

DRAFT

Quick sort and Merge Sort. **Searching**- Introduction to different searching techniques - Linear search and Binary search.		
Ex.No	Name of the Experiment	
6	Write a python program to implement bubble sort	12
7	Write a python program to implement linear search	
8	Write a python program to implement binary search	
TOTAL HOURS		**75**

Reference

1. Data Structures and Algorithms using Python by Rance D. Necaise
2. Python Data Structures and Algorithms by Benjamin Baka
3. Data Structures and Algorithms in Python by Michael T. Goodrich

Web-based/Online Resources

1. www.geeksforgeeks.com
2. www.simplilearn.com/tutorials/python-tutorial/data-structures
3. www.tutorialspoint.com/python_data_structure/index.htm

Suggested List of Students Activity

- Presentation/Seminars by students on any recent technological developments based on the course
- Periodic class quizzes conducted on a weekly / fortnightly based on the course
- Blended learning activities to explore the recent trends and developments in the field.

Equipment / Facilities required to conduct the practical portion

1. **Hardware Requirement:**
   - Desktop Computers / Laptop
   - Printer
2. **Software Requirement:**
   - Windows / Linux Operating System
   - Python IDLE / Spyder.

Continuous Assessment Pattern (CA1 & CA2)

Theory (50)

Allocation of Marks for CA1 & CA2 Theory Tests

DRAFT

Section	Description	Marks	
Part – A	11 questions to be answered out of 12	11 x 2 marks	22
Part – B	4 questions to be answered out of 6	4 x 7 marks	28
TOTAL			**50**

Practical (50)

Allocation of Marks for CA1 & CA2 Practical Tests

Section	Description	Marks
1	Aim & Procedure	35
2	Execution and Result	15
TOTAL		**50**

1052234340	JAVA PROGRAMMING	L	T	P	C
Practicum		2	0	4	4

Introduction

Java is a class-based, object-oriented programming language .It is intended to let application developers write once, and run anywhere (WORA), meaning that compiled Java code can run on all platforms that support Java without the need for recompilation. Java is widely used for developing applications for desktop, web, and mobile devices. Java is known for its simplicity, robustness, and security features, making it a popular choice for enterprise-level applications. Students will learn Java tokens, variables, data types, control structures, functions, arrays, strings, object - oriented programming concepts and swing components. Through hands-on students will develop proficiency in writing structured and efficient Java programs to solve a variety of computational problems.

Course Objectives

The objectives of this course are enabling the students

- To understand the concepts of Object Oriented Programming.

DRAFT

- To learn about the control structures, class with attributes and methods used in Java.
- To gain knowledge of arrays and strings.
- To understand the concept of exception handling mechanism.
- To comprehend the basics of swing components and its importance in application development.

Course Outcomes

At the end of the course, students will be able

CO1: Demonstrate knowledge on Java Programming fundamentals.

CO2: Develop programs in Java using control structures, array and string.

CO3: Demonstrate use of object - oriented programming concepts in Java.

CO4: Apply programming skills to solve overriding problems using interface.

CO5: Develop applications using swing components.

Pre-requisites: Nil

DRAFT

CO/PO Mapping

CO / PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	3	3	3	1	1	2
CO2	3	3	3	3	1	1	2
CO3	3	3	3	3	1	2	2
CO4	3	3	3	3	1	2	2
CO5	3	3	3	3	1	3	2

Legend: 3-High Correlation, 2-Medium Correlation, 1-Low Correlation

Instructional Strategy

- Engage and Motivate: Instructors should actively engage students to boost their learning confidence.

- Real-World Relevance: Incorporate relatable, real-life examples and engineering applications to help students understand and appreciate course concepts.
- Interactive Learning: Utilize demonstrations and plan interactive student activities for an engaging learning experience.
- Application-Based Learning: Employ a theory-demonstrate-practice-activity strategy throughout the course to ensure outcome-driven learning and employability.
- Encourage Critical Analysis: Foster an environment where students can honestly assess experiment outcomes and analyze potential sources of error in case of discrepancies.

DRAFT

Assessment Methodology

	Continuous Assessment (40 marks)					End Semester Examination (60 marks)
	CA1		CA2		CA3	
Mode	Written Test (First 50% of Theory)	Practical Test (First 50% of Exercise)	Written Test (Next 50% of Theory)	Practical Test (Next 50% of Exercise)	Model Theory Exam	Theory Exam
Duration	2 hours	2 hours	2 hours	2 hours	3 hours	3 hours
Exam Marks	50	50	50	50	100	100

Converted to	20	20	10	60
Practical log book / Observation Note book with bonafide	10			
Final Marks	40			60

Note:

- Continuous Assessment tests (CA1 and CA2) should be conducted for both Theory and Practical portions. **Best of one** will be considered for 20 marks for internal assessment.
- Continuous Assessment Model Theory examination (CA3) should be conducted as per the end semester question pattern after completion of all the portions. The marks awarded shall be converted to 10 Marks.
- Practical log book / Observation note book with bonafide carries 10 Marks.

DRAFT

1052234340	**JAVA PROGRAMMING**	**L**	**T**	**P**	**C**
Practicum		**2**	**0**	**4**	**4**

Unit I	**INTRODUCTION TO JAVA**	
Introduction to OOPS: Paradigms of Programming Languages – Basic concepts of Object-Oriented Programming –Benefits of OOPs –.– Java features – Java Environment – JDK – API. Creating and Executing a Java program – Java Tokens- Java Virtual Machine (JVM) –Command Line Arguments – Constants – Variables – Data types - Scope of variables – Type casting – Operators.		6

Ex No 1: Write a java program to read the temperature in Celsius and convert into Fahrenheit. **Ex No 2**:Write a program to read 2 integers and find the largest number using conditional operator. **Ex No 3**: Write a Java program to implement command line arguments.	12
Unit II	**CONTROL STRUCTURES, ARRAY AND STRING**
Control structures: Decision making statements - looping statements - branching statement - Arrays: One Dimensional Array –Multidimensional Array – String: String Array – String Methods.	6
Ex No 4: Write a Java program to find the sum and average of your tenth standard marks. **Ex No 5:** Write a Java Program to sort 10 student names in alphabetical order using bubble sort	12
Unit III	**CLASS AND OBJECTS**
Class and objects: Defining a class – Methods – Creating objects – Accessing class members – Constructors – Method overloading – Static members – Nesting of Methods - Final methods.	6
Ex No 6 : Write a Java program to collect student details using constructors. **Ex No 7:**Write a Java program to calculate area of rectangle, triangle and square using method overloading.	12
UNIT IV	**INHERITANCE AND INTERFACE**
Inheritance: Defining Inheritance –Types of Inheritances– Overriding Methods – Final Variables and Methods - Abstract Class**- Interfaces:** Defining Interface – Types of Interfaces.	6
Ex No 8: Write a Java program to create a class called Shape with methods called getPerimeter() and getArea(). Create a subclass called Circle that overrides the getPerimeter() and getArea() methods to calculate the area and perimeter of a circle. **Ex No 9**: Write a Java program to create an interface Shape with the getArea() method. Create three classes Rectangle, Circle, and Triangle that implement the Shape interface. Implement the getArea() method for	12

DRAFT

	each of the three classes.	
UNIT V	**EXCEPTION HANDLING AND SWING**	
Exception Handling: Basics of Exception Handling – try blocks – throwing an exception – catching an exception – finally statement. Swing Components and Event Handlers: – Event Handlers – Event Listeners –Input Events.		6
Ex No 10: Write a Java program to create a panel with three buttons, labeled `Red`, `Blue` and `Yellow`, so that clicking each button results in the background color changing to the appropriate color.		12
TOTAL HOURS		**75**

TEXT BOOKS

1. E. Balagurusamy, "Programming with Java", TataMc-Graw Hill, 5th Edition.
2. Sagayaraj, Denis, Karthick and Gajalakshmi, "Java Programming for Core and advanced learners", Universities Press (INDIA) Private Limited 2018.

REFERENCE BOOKS

1. Herbert Schildt, "The complete reference Java", TataMc-Graw Hill, 7th Edition.
2. Cay S. Horstmann, Gary Cornell, "Core Java Volume I -Fundamentals", Pearson India Education Services Pvt. Ltd.
3. Ken Arnold, James Gosling, David Holmes, "The Java™ Programming Language", Addison Wesley.

BASED/ONLINE RESOURCES

- https://www.w3schools.com/java/
- https://www.geeksforgeeks.org/java/
- https://www.tutorialspoint.com/java/index.htm
- NPTEL & MOOC courses titled Java
- https://nptel.ac.in/courses/106105191/

Suggested List of Students Activity

- Presentation/Seminars by students on any recent technological developments based on the course.
- Programming assignments
- Periodic class/online quizzes conducted based on the course.

DRAFT

- Blended learning activities to explore the recent trends and developments in the field.

Equipment / Facilities required to conduct the Practical Portion

1. **Hardware(s) Requirement:**
    - Desktop / Laptop
    - Printer
2. **Software(s) Requirement:**
    - Windows Operating System
    - Net Beans 8.0.2 / 8.2 with JDK.

Continuous Assessment Pattern (CA1 & CA2)

Theory (50)

Allocation of Marks for CA1 & CA2 Theory Tests

Section	Description	Marks	
Part – A	11 questions to be answered out of 12	11 x 2 marks	22
Part – B	4 questions to be answered out of 6	4 x 7 marks	28
TOTAL			**50**

Practical (50)

Allocation of Marks for CA1 & CA2 Practical Tests

Section	Description	Marks
1	Aim & Procedure	35
2	Execution and Result	15
TOTAL		**50**

Board Practical Examination

PART – A

1. Write a Java program to read the temperature in Celsius and convert into Fahrenheit.
2. Write a Java program to read 2 integers and find the largest number using conditional operator .
3. Write a Java program to implement command line arguments.

DRAFT

4. Write a Java program to find the sum and average of your tenth standard marks.
5. Write a Java Program to sort 10 student names in alphabetical order using bubble sort.

PART – B

6. Write a Java program to collect student details using constructors.
7. Write a Java program to calculate area of rectangle, triangle and square using method overloading.
8. Write a Java program to create a class called Shape with methods called getPerimeter() and getArea(). Create a subclass called Circle that overrides the getPerimeter() and getArea() methods to calculate the area and perimeter of a circle.
9. Write a Java program to create an interface Shape with the getArea() method. Create three classes Rectangle, Circle, and Triangle that implement the Shape interface. Implement the getArea() method for each of the three classes.
10. Write a Java program to create a panel with three buttons, labeled `Red`, `Blue` and `Yellow`, so that clicking each button results in the background color changing to the appropriate color.

SCHEME OF VALUATION		
SNO	**ALLOCATION**	**MARKS**
1	Aim (05) ,Program from Part – A (30)	35
2	Aim (05) ,Program from Part – B (30)	35
3	Executing any one program (Part A or Part –B)	15
4	Output	10
5	Viva Voce	05
6	Total	100

DRAFT

1052234440	Python Programming	L	T	P	C
Practicum		1	0	4	3

Introduction

Being able to implement the basic logical statements in python and explore python's various data structures and packages which are much useful in the fields of data science, artificial intelligence.

Course Objectives

The objective of this course is to enable the student to

1. To read and write simple python programs.
2. To define strings in python and operations on string.
3. Represent compound data using python lists, tuples, dictionaries.
4. To define and access multi-dimensional arrays using NumPy.
5. To do input/output with files in python.

Course Outcomes

After successful completion of this course, the students should be able to

CO1: Demonstrate the installation process of python IDE and modules.

CO2: Implement the decision making and looping statements in python.

CO3: Define regular expression for the pattern and verify for the validity.

CO4: Create and access string, list, tuple, dictionary and NumPy array.

CO5: Read and write text and csv file using python.

Pre-requisites: Nil

CO/PO Mapping

CO / PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	2	2	3	---	---	---
CO2	3	3	3	3	---	---	---
CO3	3	3	3	3	---	---	---
CO4	3	2	3	3	---	---	---
CO5	3	3	3	3	---	---	---

Legend: 3-High Correlation, 2-Medium Correlation, 1-Low Correlation

DRAFT

Instructional Strategy

- Engage and Motivate: Instructors should actively engage students to boost their learning confidence.
- Real-World Relevance: Incorporate relatable, real-life examples and engineering applications to help students understand and appreciate course concepts.
- Interactive Learning: Utilize demonstrations and plan interactive student activities for an engaging learning experience.
- Application-Based Learning: Employ a theory-demonstrate-practice-activity strategy throughout the course to ensure outcome-driven learning and employability.
- Encourage Critical Analysis: Foster an environment where students can honestly assess experiment outcomes and analyze potential sources of error in case of discrepancies.

Assessment Methodology

	Continuous Assessment (40 marks)					End Semester Examination (60 Marks)
	CA1		CA2		CA3	
Mode	Written Test (First 50% of Theory)	Practical Test (50% of Exercise)	Written Test (Next 50% of Theory)	Practical Test (50% of Exercise)	Model Practical Exam	Practical Exam
Duration	2 hours	2 hours	2 hours	2 hours	3 hours	3 hours
Exam Marks	50	50	50	50	100	100
Converted to	20		20		10	60
Practical log book / Observation Note book with bonafide	10					
Final Marks	40					60

Note:

DRAFT

- Continuous Assessment tests (CA1 and CA2) should be conducted for both Theory and Practical portions. **Best of one** will be considered for 20 marks for internal assessment.
- Continuous Assessment Model practical examination (CA3) should be conducted as per the end semester question pattern after completion of all the portions. The marks awarded shall be converted to 10 Marks.
- Practical logbook / Observation notebook with Bonafide carries 10 Marks.

1052234440	**Python Programming**	L	T	P	C
Practicum		1	0	4	3

Unit I	**INTRODUCTION TO PYTHON**	
Installing and running Python in interpreter and Interactive mode, Basic Datatypes in Python: int, float, string. Storing Values in Variables, Basic functions in Python: input (), print (), str (), int (), float (). Decision Making – Simple if, if…else and if … elif statement; Control Statement: while, break, continue, for loop, range ().		4
Ex No 1: Write a python program to read three numbers and print the greatest of three numbers. **Ex No 2**: Write a python program to find the sum of N number using range () function in for loop.		12
Unit II	**STRING, LIST, TUPLE, DICTIONARY**	
Sequence Datatypes. Operations on sequence datatypes: Indexing and slicing, concatenation, and replication, in and not in operators to access elements. List: Creation, mutable property, Inbuild methods of List: index (), append (), insert (), sort (), reverse (). Tuple: immutable property, converting types using tuple (), list (). Dictionary Datatype: Creation, keys (), values () and items () methods.		4
Ex No 3: Write a python program to demonstrate the string slicing, concatenation, replication and len() method. **Ex No 4:** Write a python program to create a tuple and convert into a list and print the list in sorted order. **Ex No 5:** Write a python program to create a dictionary and check whether a key or value exist in the dictionary.		16
Unit III	**NumPy**	

DRAFT

Install and import NumPy module, Creation of one dimensional, 2D-dimensional NumPy array using array (), Slicing, indexing, NumPy methods: shape (), reshape (), concatenate (), where (). Arithmetic operations in NumPy, Aggregation functions in NumPy.	4
Ex No 6: Write a python program to create one dimensional array and convert into a 2D-dimensional array using reshape(), print the first two columns alone using slicing. **Ex No 7:** Write a python program to create two-dimensional array and search for an element using where () function. **Ex No 8**: Write a python program to create a 2D-dimensional array and demonstrate aggregation functions sum (), min () and max () in the row and column wise.	16
UNIT IV	**FILE HANDLING**
Text file handling: file opening mode, reading from a file: read(), readline(), readlines() and writing into a file: write(), writeline(). Pandas package: install and import pandas, read and write a csv file, pandas methods: head(), describe().	3
Ex No 9: Write a python program to read a text file and write the content in another file. **Ex No 10**: Write a python program to read a csv file using pandas and print the content.	16
TOTAL PERIODS	**75**

Suggested List of Students Activity

- Presentation/Seminars by students on any recent technological developments based on the course.
- Periodic class/online quizzes conducted based on the course.
- Blended learning activities to explore the recent trends and developments in the field.

References

1 Automate the Boring Stuff with Python by Al Sweigart.

2 Python Data Science Handbook, Essential tool for working with data by Jake Vanderplas.

DRAFT

3 Problem Solving and Python Programming, E Balagurusamy, McGraw Hill Education, 2017 & First Edition.

4 Python for Data Analysis: Data Wrangling with Pandas, NumPy and Ipython, Wes McKinney, O'REILLY, Third Edition,2023

Web-based/Online Resources

1. NumPy : https://www.w3schools.com/python/numpy/
2. File Handling in Python: https://www.geeksforgeeks.org/file-handling-python/

   Working with Excel files using Pandas: https://www.geeksforgeeks.org/working-with-excel-files-using-pandas/

Equipment / Facilities required to conduct the Practical Portion

1. **Hardware Requirement:**
   - Desktop Computers / Laptop
   - Printer
2. **Software Requirement:**
   - Windows / Linux Operating System
   - Python IDLE /Spyder.

Continuous Assessment Pattern (CA1 & CA2)

Theory (50)

Allocation of Marks for CA1 & CA2 Theory Tests

Section	Description	Marks	
Part – A	11 questions to be answered out of 12	11 x 2 marks	22
Part – B	4 questions to be answered out of 6	4 x 7 marks	28
TOTAL			**50**

Practical (50)

Allocation of Marks for CA1 & CA2 Practical Tests

Section	Description	Marks
1	Aim & Program	35
2	Execution and Result	15

DRAFT

TOTAL	50

BOARD PRACTICAL EXAMINATION

PART – A

Ex No 1: Write a python program to read three numbers and print the greatest of three numbers.

Ex No 2: Write a python program to find the sum of N number using range () function in for loop.

Ex No 3: Write a python program to demonstrate the string slicing, concatenation, replication and len() method.

Ex No 4: Write a python program to create a tuple and convert into a list and print the list in sorted order.

Ex No 5: Write a python program to create a dictionary and check whether a key or value exist in the dictionary.

PART – B

Ex No 6: Write a python program to create one dimensional array and convert into a 2D-dimensional array using reshape (), print the first two columns alone using slicing.

Ex No 7: Write a python program to create two-dimensional array and search for an element using where () function.

Ex No 8: Write a python program to create a 2D-dimensional array and demonstrate aggregation functions sum (), min () and max () in the row and column wise.

Ex No 9: Write a python program to read a text file and write the content in another file.

Ex No 10: Write a python program to read **a csv file** using pandas and print the content.

SCHEME OF VALUATION		
SNO	**ALLOCATION**	**MARKS**
1	Aim (05) ,Program from Part – A (30)	35
2	Aim (05) ,Program from Part – B (30)	35

DRAFT

3	Executing any one program (Part A or Part –B)	15
4	Output	10
5	Viva Voce	05
6	Total	100

DRAFT

1052234540	E-PUBLISHING TOOLS	L	T	P	C
Practicum		1	0	4	3

Introduction:

This course provides an introductory exploration of e-publishing tools and technologies for creating and distributing digital publications. Students will learn about various e-publishing formats, tools used to create e-books, digital magazines, interactive documents and more. Through hands-on projects and practical exercises, students will gain proficiency in using popular e-publishing software and tools to design, format, and publish digital content for different devices and platforms.

Course Objectives

The objective of this course is to

1. Learn all tools and options in Text editing software.
2. Create Vector drawings using CorelDraw.
3. Learn all tools and options in Bitmapped image editing software.
4. Learn to use Layer masks, filters and blending modes in Adobe Photoshop.
5. Learn to use online publishing software like Canva, Figma.
6. Learn to use character styles, paragraph styles, text effects, frames in any page layout software like Adobe Indesign or any other equivalent open source software.

Course Outcomes

After successful completion of this course, the students should be able to

CO1: Create designs like Business Cards, Notebook wrapper and logo.

CO2: Create passport size photo by removing background.

CO3: Design a new image by blending two images using layer masking and filters.

CO4: Prepare new designs for brochures, calendar and invitations.

CO5: Learn online graphic design platform to design presentations, social media graphics, posters, flyers, infographics.

Pre-requisites

Basic Knowledge about computer and multimedia elements.

DRAFT

CO/PO Mapping

CO / PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	3	2	2	1	1	2
CO2	3	3	2	2	1	1	2
CO3	3	3	2	2	1	1	2
CO4	3	3	2	2	1	1	2
CO5	3	3	2	2	1	1	2

Legend: 3-High Correlation, 2-Medium Correlation, 1-Low Correlation

DRAFT

Instructional Strategy:

1. It is advised to assign assign hands-on projects to students create digital publications using e-publishing tools such as Adobe Photoshop, CorelDraw, Adobe PageMaker etc. Projects could include designing and formatting interactive documents.
2. Conduct of interactive demos to help students learn specific techniques and tools. Provide step-by-step guidance and encourage questions and participation.
3. Analysis of real-world examples of successful digital publications and their design, formatting, and distribution strategies.

Assessment Methodology

	Continuous Assessment (40 marks)					End Semester Examination (60 Marks)
	CA1		CA2		CA3	
Mode	Written Test (First 50% of Theory)	Practical Test (50% of Exercise)	Written Test (Next 50% of Theory)	Practical Test (50% of Exercise)	Model Practical Exam	Practical Exam
Duration	2 hours	2 hours	2 hours	2 hours	3 hours	3 hours
Exam Marks	50	50	50	50	100	100
Converted to	20		20		10	60
Practical log book / Observation Note book with bonafide	10					
Final Marks	40					60

Note:

- Continuous Assessment tests (CA1 and CA2) should be conducted for both Theory and Practical portions. **Best of one** will be considered for 20 marks for internal assessment.
- Continuous Assessment Model practical examination (CA3) should be conducted as per the end semester question pattern after completion of all the portions. The marks awarded shall be converted to 10 Marks.

DRAFT

1052234540	E-PUBLISHING TOOLS	L	T	P	C
Practicum		1	0	4	3

Unit I E-PUBLISHING & VECTOR DRAWING APPLICATIONS	
Introduction to E-Publishing- Evolution -Categories of E-Publishing Tools- Text Editors- Word Processors- Vector Drawing Applications- Page Layout Applications- Bitmapped image Applications. Installing vector drawing application - Starting and Opening Drawings - Previewing - Viewing Modes - Saving and Closing Drawings - Workspace - Lines, Shapes, and Outlines- Working with Objects, Symbols, and Layers- Colour, Fills, and Transparencies- Exploring Special Effects- Working with Text- Templates and Styles- Pages and Layout.	**6**
1. Create a business card with a logo using various text styles, rectangle tool,and ellipse tool.	**6**
2. Design a notebook wrapper using fountain filling and pattern filling tools.	**6**
3. Transform one object into another object using a blend tool. (Students can be allowed to use any other open source vector drawing software)	**6**
Unit II PAGE LAYOUT APPLICATIONS	
Getting started with Scribus- Opening, closing and navigating- Text Tools- Shape Tools- Image Frame Tools- Color Management Tools- Master Pages- Layers- Alignment and Distribution Tools- PDF Export Options.	**2**
4. Design an invitation for your college convocation using text tools Shape Tools- Image tables in the page layout software. (Students can be allowed to use any other open source page layout software)	8
Unit III BITMAPPED IMAGE APPLICATIONS	
Installing Image editing application- Opening, moving, editing, saving images-	**4**

DRAFT

Essential Keyboards Shortcuts- Workspace- panels- Selection tools- Crop and slice tools- Colour selection and measuring tools- Text tools- Navigation tools- Retouching tools- Painting tools- Drawing tools- Customizing Toolbars- Layers - Layer Mask- Blending modes- Filters.	
5.Create a design by using various selection tools, cutting and pasting the images.	**6**
6. Create a passport size photo by removing the background of a photo and change it to blue color.	**6**
7. Change the image looks by applying various filters and blending modes. (any one among Pencil sketch, Water Color, Blurred Background)	**6**
8. Create a design with the use of a layer mask using two images as background and foreground. (Students can be allowed to use any other open source image editing software)	**6**
Unit IV ONLINE GRAPHIC DESIGN (CANVA CONTROLS)	
Introduction to Canva- Templates- Backgrounds- Working with text- Font Styles- Elements- images, icons, or graphs- Shapes- Audio- Video- Animation- Applying Filters and Effects- Save- Download and share.	**3**
9. Design a multipage document like a tri-fold brochure using various elements for the college workshop.	**5**
10.Prepare a stylish calendar sheet by using tables and its formatting tools. (Students can be allowed to use any other open source online graphic design software)	**5**
Total Periods	**75**

DRAFT

Text and Reference Books:

1. “Gary David Bouton” "CorelDraw X7: The Official Guide- 11th Edition", O’Reilly Media.
2. ”Conrad Chavez, Andrew Faulkner” “Adobe photoshop classroom in a book” Adobe Press.
3. “Peter Bauer” “Adobe photoshop CC for Dummies’’ John Wiley & Sons.
4. “Roger Wambolt” "CorelDraw Graphics Suite 2021: The Ultimate Step-by-Step Guide to Learn CorelDraw Graphics Suite"

5. “Patrick Ejeke ” “Canva: Professional Tips and Tricks When You Design with Canva”
6. “Robert Lewis” “Canva for Beginners and Business”

Web-based/Online Resources:

- https://www.psdstack.com/resources/photoshop-tutorials/
- https://www.vandelaydesign.com/free-CorelDraw-tutorial
- https://www.canva.com/designschool/tutorials/
- https://www.youtube.com/watch?v=uCcPDSE6vLw
- https://www.youtube.com/watch?v=j4y6S20yx7o
- https://www.youtube.com/watch?v=cLI52T0OmtY
- https://www.scribd.com/doc/13080717/CorelDraw-Course-Manual
- Getting Started with Adobe Photoshop (photoshopessentials.com)
- https://www.CorelDraw.com/en/learn/tutorials/
- https://www.CorelDraw.com/en/learn/webinars/
- Create Wedding Invitation Cards in Illustrator: Start to Finish! (youtube.com)

Suggested List of Students Activity:

- Presentation/Seminars by students on any recent technological developments based on the course
- Periodic class quizzes conducted on a weekly/fortnightly based on the course
- Micro project that shall be an extension of any practical lab exercise to real-world application.

Equipment / Facilities required to conduct the Practical Course

HARDWARE REQUIREMENTS

Desktop Computers with Internet Connectivity , Laser printer , Scanner

SOFTWARE REQUIREMENTS

Any Open Source Software , GIMP , Scribus , Inkscape ,Adobe Photoshop ,CorelDraw

Continuous Assessment Pattern (CA1 & CA2)

Theory (50)

Allocation of Marks for CA1 & CA2 Theory Tests

Section	Description	Marks

DRAFT

Part – A	11 questions to be answered out of 12	11 x 2 marks	22
Part – B	4 questions to be answered out of 6	4 x 7 marks	28
TOTAL			**50**

Practical (50)

Allocation of Marks for CA1 & CA2 Practical Tests

Section	Description	Marks
1	Aim & Program	35
2	Execution and Result	15
TOTAL		**50**

BOARD PRACTICAL EXAMINATION

PART - A

1. Create a business card with a logo using various text styles, rectangle tool, and ellipse tool.
2. Design a notebook wrapper using fountain filling and pattern filling tools.
3. Transform one object into another object using a blend tool.
4. Design an invitation for your college convocation using text tools Shape Tools- Image tables in the page layout software.
5. Create a design by using various selection tools, cutting and pasting the images.

PART - B

6. Create a passport size photo by removing the background of a photo and change it to blue color.
7. Change the image looks by applying various filters and blending modes. (any one among Pencil sketch, Water Color, Blurred Background)
8. Create a design with the use of a layer mask using two images as background and foreground.
9. Design a multipage document like a tri-fold brochure using various elements for the college workshop.
10. Prepare a stylish calendar sheet by using tables and its formatting tools.

DRAFT

SCHEME OF VALUATION		
SNO	**ALLOCATION**	**MARKS**
1	Aim (05) ,Program from Part – A (30)	35
2	Aim (05) ,Program from Part – B (30)	35
3	Executing any one program (Part A or Part –B)	15
4	Output	10
5	Viva Voce	05
6	Total	100

DRAFT

1052234652	SCRIPTING LANGUAGES	L	T	P	C
Project		0	0	6	2

Rationale:

The main objective of the course is to introduce the students with the advanced Web-based software development using JavaScript, PHP, and MySQL. The subject will impart knowledge to design visually appealing, dynamic, device-independent, and interactive web-based applications with client-side and server-side scripting. Additionally, this course aims at developing innovative skills in the students whereby they apply the knowledge and skills gained through the course by undertaking a project. The individual students have different skills, attitudes, and strengths. At the end of this course, the students will learn how to work with the team and how to prepare the report.

Course Objectives:

- To learn to utilize the PHP statements for Application Development
- To learn to develop web applications using PHP and MysQL
- To design the interactive and dynamic web applications using AJAX,JQUERY and Node.js
- To learn to work in teams and to utilize the knowledge gained into an application suitable for a real practical working environment
- Learn and understand the gap between the technological knowledge acquired and the actual industrial need and to compensate it by acquiring additional knowledge as required

Course Outcomes:

At the end of this course, students will be able to:

CO1: Write PHP script to store, access and display the data in the MySQL Database.

CO2: Design the dynamic web pages using AJAX, Jquery and Node.js

CO3: Develop device-independent Web application using Bootstrap

CO4: Identify, discuss and justify the technical aspects of the chosen project with a Comprehensive and systematic approach with the team.

CO5: Communicate and report effectively project related activities and findings and reproduce, improve and refine the acquired result

Pre-requisites:-

DRAFT

CO/PO Mapping

CO / PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	3	3	2	1	-	1
CO2	3	3	3	2	1	-	1
CO3	3	3	3	3	2	-	2
CO4	3	3	3	3	2	3	3
CO5	3	3	3	3	2	3	3

Legend: 3-High Correlation, 2-Medium Correlation, 1-Low Correlation

Instructional Strategy:

- Real-World Relevance: Incorporate relatable, real-life examples and engineering applications to help students understand and appreciate course concepts.
- Interactive Learning: Utilize demonstrations and plan interactive student activities for an engaging learning experience.
- Simulation and Real-World Practice: Conduct demonstrations and hands-on activities in with built in Models
- Encourage Critical Analysis and Thinking: Foster an environment where students can think over the real world problem and find the solution for the same also they can honestly assess experiment outcomes and analyze potential sources of error in case of discrepancies.
- Cooperative & Team-Based Learning: Foster the environment where in the students can work in the team, discuss among the team to find the solution for the real-world problem.

Guidelines for Project Team Formulation

DRAFT

Batch size: Maximum 6 students per batch

Assessment Methodology:

	Continuous Assessment (40 marks)				End Semester Examination (60 marks)
	CA1	CA2	CA3		
Mode	Practical Test (First 50% of Exercise)	Practical Test (Next 50% of Exercise)	Review -1 (after 6th Week)	Review - 2 (after 12th Week)	
Duration	2 hours	2 hours	2 hours	2 hours	3 hours
Exam Marks	50	50	50	50	100
Converted to	20	20	10	10	60
Practical log book / Observation Note book with bonafide	10				
Final Marks	40				60

Note:

- CA1 and CA2 Assessment tests should be conducted for the Practical portions. . **Best of one** will be considered for 20 marks for internal assessment.
- Two reviews should be conducted, each for 50 Marks, and it is converted to 10 Marks. The sum of both the review marks is considered for CA3. Proper records should be maintained for the two Project Reviews. Each review is evaluated as per the following guidelines:

Details of Mark allocation	Max. Marks

DRAFT

1052234652	SCRIPTING LANGUAGES	L	T	P	C
Project		0	0	6	2

Presentation	20
Implementation	20
Viva Voce	10
Total	50

Unit I	PHP INTRODUCTION	
Theory: Introduction to Server Side Scripting - PHP: PHP Structure and Syntax - Integrating HTML with PHP - if Statements - if and else – switch case - for loop – for each loop- Strings – Arrays - HTML Form Elements Processing in PHP - Passing Variables between Pages.		12
Practice Experiments: **Ex No 1:** Write PHP code to implement any five string and array functions **Ex No 2:**Design the HTML form to collect student biodata and SSLC Mark, Process the collected data in the PHP and Find Total and Average for Mark.		
Unit II	**PHP PRGRAMMING AND MYSQL PHP**	
Theory: Sessions and Cookies- Page redirection- Connecting to the MySQL Server – Insert, Edit, Update, Delete and Querying the Database from PHP		12
Practice Experiments: **Ex No 3:**Develop the simple application which display result of the student by getting register number as user input(assume student marks are already available in the database) **Ex No 4:** Develop the simple login page, which validates the username, and password (assume username, password and student_name are stored in the database). If username and password are correct, the page should redirect to Welcome.phpfile and display the student_name in that page. If username or password is incorrect page should remain in login page itself.		
Unit III	**AJAX AND JQUERY**	
Theory: Introduction to AJAX -The XMLHttpRequest Object - JSON - Introduction to		12

DRAFT

jQuery - jQuery Events – jQuery Effects - AJAX and JQuery.		
Practice Experiments: **Ex No 5:**Write the code to disable right-click option in the webpage using the jQuery **Ex No 6:** Develop the simple application which display details of the collegeby getting college code as input using AJAX without reloading the page (assume college details like code, name, courses_offered, address, hostel facility,etc., are already available in the database)		
Unit IV	**WEB APPLICATIONS FRAMEWORKS**	
Theory: Bootstrap 5.0: Cards – Nav Bar- Form elements- Node.js : Introduction – NPM-Node js Modules-upload files- Send an Email – Events-Node.js and Mysql-introduction to Django		12
Practice Experiments: **Ex No 7:**Develop the Node.js code to upload the file to server **Ex No 8:**Develop the Node.js code to send an email		
	Project Development & Report Preparation	42
	TOTAL PERIODS	**90**

Suggested List of Project (Not limited to this list):

1. College Management Software
2. Livestock Management software
3. Online Student Attendance and Biodata Management Software
4. Online Employee Management Software
5. Online Event Management Software
6. Online Transport Management Software
7. Online Library Management Software
8. Online Blood Bank
9. Online Shopping / Billing Software
10. Online Book / Music Store

Suggested List of Students Activity:

- Presentation/Seminars by students on any recent technological developments in Web development.
- online quizzes
- Project Development

DRAFT

Text Books:

1. Thomas Powell, Fritz Schneider "Java Script: The Complete Reference", Tata McGras-Hill, Third Edition, July 2017.
2. Timothy Boronczyk, Elizabeth Naramore, Jason Gerner, Yann Le Scouarnec, Jeremy Stolz, Michael K. Glass "Beginning PHP6, Apache, MySQL, Web Development", Wiley Pub., 2009.
3. David Barron: "The World of Scripting Languages", 1st Edition, Wiley publications

Reference Books:

1. Thomas Powell "Ajax: The Complete Reference ", 2008,Mc grew hill.
2. SandroPasquali, Kevin Faaborg "Mastering Node.js" Second Edition, December 2017,Packt Publishing
3. Ben Shaw, SaurabhBadhwar, Andrew Bird, Bharath Chandra K S, Chris Guest "Web Development with Django",· 2021
4. Alex Libby, "Mastering jQuery", Packet Publications first edition,2015
5. AzatMardan, "Practical Node. Js Building Real-World Scalable Web Apps", APRESS Publication, 2018

Web-based/Online Resources:

1. *https://archive.org/details/BeginningPHP6ApacheMySQL6WebDevelopment/page/n7/mode/2up*
2. *https://www.tutorialspoint.com/nodejs/nodejs_introduction.htm*
3. *https://www.tutorialspoint.com/php/index.htm*
4. *https://www.w3schools.com/nodejs/*
5. *https://www.w3schools.com/jquery/default.asp*
6. *https://www.w3schools.com/php/default.asp*
7. *https://phpgurukul.com*

Equipment / Facilities required conducting the Practical Course / Project

Hardware Required.

1. Desktop / Laptop Computers.

Software Required.

1. Apache / Httpd / Wamp/ Xamp Webserver
2. MySQL
3. Any Web browser

BOARD PRACTICAL EXAMINATION:

DRAFT

SCHEME OF VALUATION		
SNO	**ALLOCATION**	**MARKS**
1	Implementation /Demo	50
2	Report	30
3	Viva Voce	20
4	Total	100

DRAFT

1052235130	Cloud Computing	L	T	P	C
Practicum		2	0	2	3

Rationale

This course will introduce the emerging technology cloud computing. The advantages of cloud services and cloud applications will give the students the much needed exposure to the current trend.

Course Objectives

The objective of this course is to

1. Introduce the concept of virtualization.
2. Outline the concepts of cloud computing.
3. Summarize the different types of web services and cloud service providers.
4. Elaborate the security issues in cloud.
5. Present the cloud applications.

Course Outcomes

After successful completion of this course, the students should be able to

CO1: Understand the concept of virtualization.

CO2: Differentiate various cloud services.

CO3: Explore the different cloud service providers.

CO4: Understand the security issues with cloud and security policies.

CO5: Comprehend the various cloud applications.

Pre-requisites

The student should have taken up Computer Networks and Security

The student should have the basic Knowledge about network protocols.

CO/PO Mapping

CO / PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	3	2	1	1	1	3
CO2	2	3	3	1	1	1	3
CO3	3	2	2	1	1	1	3
CO4	3	2	3	1	1	1	3
CO5	3	2	3	2	1	1	3

DRAFT

Legend: 3-High Correlation, 2-Medium Correlation, 1-Low Correlation

Instructional Strategy

- The teacher can use experiential learning as an instructional strategy both in and outside the classroom.
- It may be necessary for the teacher to pre-teach the skills and processes necessary to achieve the intended learning outcomes.
- The teacher needs to encourage students to share their thoughts so that the entire class can benefit from individual insights.
- Teachers can encourage divergent thinking by asking students to transform a teacher guided image into several others of their own creation.

Assessment Methodology

	Continuous Assessment (40 marks)					End Semester Examination
	CA1		CA2		CA3	
Mode	Written Test (First 50% of Theory)	Practical Test (50% of Exercise)	Written Test (Next 50% of Theory)	Practical Test (50% of Exercise)	Model Practical Exam	Theory Exam
Duration	2 hours	2 hours	2 hours	2 hours	3 hours	3 hours
Exam Marks	50	50	50	50	100	100
Converted to	20		20		10	60
Practical log book / Observation Note book with Bonafide	10					
Final Marks	40					60

Note:

- Continuous Assessment tests (CA1 and CA2) should be conducted for both Theory and Practical portions. **Best of one** will be considered for 20 marks for internal assessment.

DRAFT

- Continuous Assessment Model Theory examination (CA3) should be conducted as per the end semester question pattern after completion of all the portions. The marks awarded shall be converted to 10 Marks.
- Practical logbook / Observation notebook with bonafide carries 10 Marks.

1052235130	**Cloud Computing**	L	T	P	C
Practicum		2	0	2	3

Unit - 1	**CLOUD COMPUTING**	
Characteristics of Cloud - Benefits – limitations - Cloud Deployment Models - Cloud service models - Infrastructure as a service(IaaS), Platform as a Service(PaaS), Software as a Service (SaaS), Anything as a Service(XaaS) – SPI Vs. Traditional IT Models – Cloud Data Centers.		6
Ex.No.	Name of the Experiment	
1	Create a professional portfolio using Google Slides.	6
2	Use Google Drive as Storage as a Service to Store, Organize, Share and Collaborate.	
Unit - 2	**VIRTUALIZATION**	
Virtualization, Hypervisors, Types of hypervisors. Virtualization techniques - para virtualization - full virtualization, - hardware assisted virtualization - hybrid virtualization.		6
Ex.No.	Name of the Experiment	
3	Install Virtualbox/VMware Workstation with different flavours of linux or windows OS on top of existing OS.	6
4	Setting up a Java development Virtual Machine with VirtualBox.	
Unit - 3	**WEB SERVICES AND CLOUD SERVICE PROVIDERS**	
Web Services and its types, Google compute engine, Google App Engine, Amazon - Amazon Elastic Compute Cloud - Amazon Simple DB - Amazon Simple Storage Service (S3) - Amazon Cloud Front.		6
Ex. No.	Name of the Experiment	
5	Install a PaaS Engine (Google App Engine / AWS / Azure or any one	6

DRAFT

	PaaS Engine) and create simple web applications using python/java.	
Unit - 4	**SECURITY IN THE CLOUD**	
Storage Location and Tenancy -Cloud Security Challenges -CSA Reference Model - Security Policies and Implementation – Virtualization Security Management.		6
Ex. No.	Name of the Experiment	
6	Protect Google Sheets and Range with various access permissions.	6
Unit - 5	**CLOUD COMPUTING APPLICATIONS**	
Adobe Creative Cloud Firefly design models– Git Hub repository basics– LinkedIn Benefits– Zoho Work Drive Features – Storage Concept of Google Big table.		6
Ex. No.	Name of the Experiment	
7	Setup a GitHub account and develop a simple python application.	6
8	Create a column family with an age-based garbage-collection policy in Google Bigtable.	
TOTAL PERIODS		**60**

Reference

- Cloud Computing: Implementation, Management and Security, Rittinghouse, John W., and James F. Ransome, CRC Press, 2017.
- Building Cloud and Virtualization Infrastructure, Lavanya S, Venkatachalam K, Saravanakumar N M, Balamurugan S,BPB Publications, 2021, 9789390684472.
- Rajkumar Buyya, Christian Vecchiola, S. Thamarai Selvi, ―Mastering Cloud Computing‖, Tata McGraw Hill, 2013.
- Toby Velte, Anthony Velte, Robert Elsenpeter, "Cloud Computing - A Practical Approach‖, Tata Mcgraw Hill, 2009.Cloud Application Architectures: Building Applications and Infrastructure in the Cloud: Transactional Systems for EC2 and Beyond (Theory in Practice), George Reese, O'Reilly, 2009.

Web-based/Online Resources

- https://www.virtualbox.org/
- https://colab.research.google.com/

DRAFT

- https://cloud.google.com/appengine/
- http://www.seanmcilvenna.com/2018/03/26/setting-up-a-java-development-vm-with-virtualbox/
- https://cloud.google.com/bigtable/docs/samples/bigtable-create-family-gc-max-age?hl=en
- https://www.accenture.com/gb-en/case-studies/about/cloud-security
- https://www.zoho.com/workdrive/features.html

Suggested List of Students Activity

- Presentation/Seminars by students on any recent technological developments based on the course
- Micro project that shall be an extension of any practical lab exercise to real-world application

Continuous Assessment Pattern (CA1 & CA2)

Theory (50)

Allocation of Marks for CA1 & CA2 Theory Tests

Section	Description	Marks	
Part – A	11 questions to be answered out of 12	11 x 2 marks	22
Part – B	4 questions to be answered out of 6	4 x 7 marks	28
TOTAL			**50**

Practical (50)

Allocation of Marks for CA1 & CA2 Practical Tests

Section	Description	Marks
1	Aim & Program	35
2	Execution and Result	15
TOTAL		**50**

DRAFT

1052235231	MACHINE LEARNING	L	T	P	C
Theory		3	0	0	3

Introduction:

With the increased availability of data from varied sources there has been increasing attention paid to the various data driven disciplines such as analytics and machine learning. Therefore, this course provides an introduction to the field of machine learning, covering fundamental concepts, algorithms, and applications. It provides a set of techniques that can automatically detect patterns in data which can then be utilized for predictions and for developing models.

Course Objectives:

- To learn the basic concepts of machine learning.
- To gain knowledge on supervised learning concepts and their applications.
- To understand unsupervised learning models and their applications.
- To evaluate the algorithms based on corresponding metrics identified
- To learn other learning aspects such as reinforcement learning and other technologies

Course Outcomes:

On successful completion of this course, the student will be able to

CO1: Explain the basic concepts of machine learning algorithms.

CO2: Evaluate and compare various machine learning models

CO3: Design and Develop various supervised learning models.

CO3: Design and Develop various unsupervised learning algorithms

CO5: Apply machine learning techniques to solve real-time problems

Pre-requisites: Nil

DRAFT

CO/PO Mapping

CO / PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	3	3	3	1	1	1
CO2	3	3	3	2	1	1	1
CO3	3	3	3	3	1	2	2
CO4	3	3	3	3	1	1	2
CO5	3	3	3	3	1	2	2

Legend: 3-High Correlation, 2-Medium Correlation, 1-Low Correlation

Instructional Strategy:

- **Engage and Motivate:** Instructors should actively engage students to boost their learning confidence.
- **Real-World Relevance:** Incorporate relatable, real-life examples and engineering applications to help students understand and appreciate course concepts.
- **Interactive Learning**: Utilize demonstrations and plan interactive student activities for an engaging learning experience.
- **Application-Based Learning**: Employ a theory-demonstrate-practice-activity strategy throughout the course to ensure outcome-driven learning and employability.
- **Encourage Critical Analysis:** Foster an environment where students can honestly assess experiment outcomes and analyse potential sources of error in case of discrepancies.

DRAFT

Assessment Methodology:

	Continuous Assessment (40 marks)			End Semester Examination (60 marks)
	CA1	CA2	CA3	
Mode	Written Test (Unit I & II)	Written Test (Unit III & IV)	Model Theory Exam	Theory Exam
Duration	2 Hours	2 Hours	3 Hours	3 Hours
Exam Marks	50	50	100	100
Converted to	20	20	20	60
Final Marks	40			60

Note:

- CA1 and CA2 Assessment tests should be conducted.CA1 covers unit 1 and unit 2, CA2 covers unit 3 and unit 4. Best of one will be considered for 20 Marks for internal assessment.
- CA3 Model examination should be conducted as per the end semester question pattern after completion of all the portions. The marks awarded shall be converted to 20 Marks.

DRAFT

1052235231	MACHINE LEARNING	L	T	P	C
Theory		**3**	**0**	**0**	**3**

Unit I	**INTRODUCTION TO MACHINE LEARNING**	
Fundamentals of Machine Learning (ML): Definition and Scope –-Basic Steps in Knowledge Discovery Process - Types- Applications. Data Descriptive Analysis: Mean- Median- Mode -Standard Deviation-Percentile Data. Workflow of Machine Learning Application Development: Data Cleaning, Data Integration, Data Reduction, Data Transformation, Algorithms and Visualizing Results.		9
Unit II	**PREDICTIVE MODELLING**	
Predictive Modelling: Basic Concepts- Needs- Types- Regression: Linear Regression–Logistic Regression-Evaluation Metrics for Regression. Frequent Pattern Mining: Needs-Associations Rules Mining-Algorithms: Apriori Algorithm-Pattern Evaluation Measures.		9
Unit III	**SUPERVISED LEARNING TECHNIQUES**	
Classification: Basic Concepts- Needs- Types - Features, Labels, Training Data, Testing Data, and models. Algorithms: Decision Tree Induction - Naive Bayes classifier – K-Nearest Neighbors'(KNN) - Model Evaluation Metrics. Ensemble Approaches: Voting Classifiers – Bagging and Boosting Sampling Techniques.		9
Unit IV	**UNSUPERVISED LEARNING TECHNIQUES**	
Clustering: Basic Concepts- Needs- Types- Types of Data-Data similarity and Dissimilarity Measures –Partitioning Method: K-Means Algorithms - Hierarchical Method: Agglomerative-Divisive Algorithm. Cluster Analysis: Metrics for Evaluating Clusters.		9
Unit V	**ADVANCED CONCEPTS, TRENDS AND APPLICATIONS**	
Text Mining- Web Mining- Time series Analysis. Case Studies: Market Basket Analysis- Recommendation System- Email Spam and Malware Filtering – Online Fraud Detection – Weather Forecasting- Challenges and Issues in Machine Learning- Tools for Machine Learning.		9
TOTAL PERIODS		**45**

DRAFT

Suggested List of Students Activity:

- Presentation/Seminars by students on any recent technological developments based on the course.
- Project based Learning in emerging application areas like finance, healthcare etc.
- Periodic class/online quizzes conducted based on the course.
- Blended learning activities to explore the recent trends and developments in the field.
- Assignments on different types of learnings
- Tutorials on solving problems using machine learning.
- Flipped classroom activities to explore application areas

TEXT BOOK

1. Ethem Alpaydin, "Introduction to Machine Learning", Fourth Edition, MIT Press ,2020.
2. Jiawei Han, Micheline Kamber, Jian Pei, "Data Mining: Concepts and Techniques", Morgan Kaufmann Publishers,3rd Edition, 2011.

REFERENCE BOOKS

1. Christopher M. Bishop, "Pattern Recognition and Machine Learning", Springer, 2006.
2. Tom Mitchell, "Machine Learning", McGraw Hill, 3rd Edition, 1997.
3. Mehryar Mohri, Afshin Rostamizadeh, Ameet Talwalkar, "Foundations of Machine Learning", Second Edition, MIT Press, 2012, 2018.
4. Sebastain Raschka, Vahid Mirjalili, "Python Machine Learning", Packt publishing, 3rd Edition, 2019.
5. Stephen Marsland, "Machine Learning: An Algorithmic Perspective, "Second Edition", CRC Press, 2014.

WEB-BASED/ONLINE RESOURCES:

- https://www.javatpoint.com/machine-learning

DRAFT

- https://www.kaggle.com/learn/intro-to-machine-learning
- https://www.w3schools.com/python/python_ml_getting_started.asp
- https://nptel.ac.in/courses/106106139
- https://nptel.ac.in/courses/106106236
- https://egyankosh.ac.in/

Continuous Assessment Pattern (CA1 & CA2)

Theory (50)

Allocation of Marks for CA1 & CA2 Theory Tests

Section	Description	Marks	
Part – A	11 questions to be answered out of 12	11 x 2 marks	22
Part – B	4 questions to be answered out of 6	4 x 7 marks	28
TOTAL			**50**

DRAFT

1052235212	Data Warehousing And Data Mining	L	T	P	C
Theory		3	0	0	3

Introduction

This course covers key aspects of data management and analysis. It starts with Data Warehousing, covering architecture, Dimensional Modeling, and ETL processes, along with tools and technologies. Then, students explore Data Warehousing Techniques, focusing on lifecycle management, metadata, and cloud-based analysis. The course then moves to Data Mining, covering fundamental concepts, preprocessing, classification, and advanced techniques like Support Vector Machines and Neural Networks. Real-world Applications and Case Studies demonstrate Data Mining's use in retail, healthcare, CRM, and fraud detection.

Course Objectives

The objective of this course is to enable the student to

- Learn Data Warehousing fundamentals.
- Acquaint themselves with various Data Warehousing tools and technologies.
- Understand the Data Warehousing lifecycle, emphasizing quality, metadata management, and cloud analysis.
- Explore core Data Mining concepts, preprocessing, and classification/clustering methods.
- Master advanced Data Mining techniques.
- Analyze real-world Data Mining applications in different sectors.

Course Outcomes

After successful completion of this course, the students should be able to

CO1: Understand Data Warehousing principles, architecture, and ETL processes, including Dimensional Modeling.

CO2: Utilize tools and technologies proficiently for Data Warehousing.

CO3: Develop skills in core Data Mining concepts.

CO4: Master advanced Data Mining techniques like Support Vector Machines, Neural Networks, Text Mining, and DBSCAN for in-depth analysis.

CO5: Apply Data Warehousing and Data Mining in real-world scenarios.

Pre-requisites : Nil

	Data Warehousing And Data Mining	L	T	P	C
Theory		3	0	0	3

CO/PO Mapping

CO / PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	3	3	1	1	2	1
CO2	3	3	3	1	1	3	2
CO3	3	3	3	1	1	3	1
CO4	3	3	3	1	1	3	2
CO5	3	3	3	1	1	3	2

Legend: 3-High Correlation, 2-Medium Correlation, 1-Low Correlation

Instructional Strategy

- Engage and Motivate: Instructors should actively engage students to boost their learning confidence.
- Real-World Relevance: Incorporate relatable, real-life examples and engineering applications to help students understand and appreciate course concepts.
- Interactive Learning: Utilize demonstrations and plan interactive student activities for an engaging learning experience.
- Application-Based Learning: Employ a theory-demonstrate-practice-activity strategy throughout the course to ensure outcome-driven learning and employability.
- Simulation and Real-World Practice: Conduct demonstrations and hands-on activities in a simulated environment, transitioning to real-world scenarios when possible.
- Encourage Critical Analysis: Foster an environment where students can honestly assess experiment outcomes and analyze potential sources of error in case of discrepancies.

DRAFT

Assessment Methodology

	Continuous Assessment (40 marks)			End Semester Examination (60 marks)
	CA1	CA2	CA3	
Mode	Written Test (Unit I & II)	Written Test (Unit III & IV)	Model Theory Exam	Theory Exam
Duration	2 Hours	2 Hours	3 Hours	3 Hours
Exam Marks	50	50	100	100
Converted to	20	20	20	60
Final Marks	40			60

Note:

- CA1 and CA2 Assessment tests should be conducted.CA1 covers unit 1 and unit 2, CA2 covers unit 3 and unit 4. Best of one will be considered for 20 Marks for internal assessment.
- CA3 Model examination should be conducted as per the end semester question pattern after completion of all the portions. The marks awarded shall be converted to 20 Marks.

.

DRAFT

1052235212	Data Warehousing And Data Mining	L	T	P	C
Theory		3	0	0	3

Unit I	INTRODUCTION TO DATA WARE HOUSING	
Introduction to Data Warehousing: Concepts and Architecture - Data Warehouse Design: Dimensional Modeling, Fact, and Dimension Tables - ETL Processes: Data Extraction, Transformation, and Loading - Data Warehouse Implementation: Tools and Technologies		8
Unit II	**DATA WAREHOUSING TECHNIQUES AND TOOLS**	
Data Warehouse Lifecycle: Planning, Design, Implementation, and Maintenance - Data Quality and Metadata Management - OLAP and Multidimensional Data Analysis - Data Warehousing in Cloud Environments		9
Unit III	**INTRODUCTION TO DATA MINING**	
Fundamentals of Data Mining: Concepts, Tasks, and Challenges - Data Preprocessing: Data Cleaning, Integration, Transformation, and Reduction - Classification Techniques: Decision Trees, Naive Bayes, and k-Nearest Neighbors - Clustering Techniques: K-means, Hierarchical Clustering - Association Rule Mining and Frequent Pattern Analysis		10
Unit IV	**CLASSIFICATION TECHNIQUES**	
Introduction to Classification: supervised learning and classification - Decision Trees: ID3, C4.5, and CART algorithms, handling overfitting and pruning techniques - Naive Bayes Classifier: Principles of Bayesian classification, Naive Bayes algorithm for text classification - k-Nearest Neighbors (KNN): KNN algorithm, Distance metrics and parameter selection		10
Unit V	**CLUSTERING TECHNIQUES**	
Introduction to Clustering – Basics concepts of clustering – Partitioning Methods: K-Means clustering – hierarchical clustering: Agglomerative and Divisive Hierarchical clustering Methods - Comparison between partitioning and Hierarchical clustering approaches		8
TOTAL PERIODS		**45**

DRAFT

Suggested List of Students Activity

- Presentation/Seminars by students on any recent technological developments based on the course.
- Periodic class/online quizzes conducted based on the course.
- Blended learning activities to explore the recent trends and developments in the field.

Text Book and References

- "Data Warehousing: Concepts, Techniques, Products and Applications" by S. S. V. Rajan.
- Data Warehousing: Data Mining & OLAP" by Alexis Leon and Mathews Leon.
- ”Data Mining: Techniques, Methods and Algorithms" by B. B. Priyadarsini and C. Malarvizhi”
- “Data Mining: Concepts and Techniques" by P. V. Rajkumar and K. Sarukesi”
- "Data Warehousing and Data Mining:Techniques,Tools, andApplications"by S. K. Singh and S. Dhamija.

Web-based/Online Resources

https://www.geeksforgeeks.org/data-mining/

https://www.javatpoint.com/data-mining-cluster-analysis

https://www.tutorialspoint.com/dwh/dwh_schemas.htm

Continuous Assessment Pattern (CA1 & CA2)

Theory (50)

Allocation of Marks for CA1 & CA2 Theory Tests

Section	Description	Marks	
Part – A	11 questions to be answered out of 12	11 x 2 marks	22
Part – B	4 questions to be answered out of 6	4 x 7 marks	28
TOTAL			**50**

DRAFT

DRAFT

1052235233	Ethical Hacking	L	T	P	C
Theory		3	0	0	3

Rationale :

Ethical hacking is designed to provide individuals with the knowledge and skills required to understand, identify, and mitigate security vulnerabilities and threats in computer systems, networks, and applications. This course introduces the concepts of Ethical Hacking and gives the learner the opportunity to learn about different tools and techniques in Ethical hacking and security and to identify and analyze the stages an ethical hacker requires to take in order to compromise a target system as well as will apply preventive, corrective and protective measures to safeguard the system.

Course Objectives:

- Learn the fundamentals of ethical hacking principles, methodologies, and terminology, distinguishing between ethical and malicious hacking practices.
- Learn to identify and assess vulnerabilities and weaknesses in computer systems, networks, and applications through various reconnaissance techniques.
- Explore various hacking tools and techniques used by ethical hackers.
- Learn network scanning and penetration testing to identify security flaws and assess the effectiveness of defense mechanisms.
- Understand key information security concepts and their relevance to ethical hacking.
- Explore common attack vectors and learn how to defend against them.
- Learn how to secure systems and networks by implementing intrusion detection and prevention systems, firewalls, and encryption.

Course Outcomes:

On successful completion of this course, the student will be able to

CO1: Students will gain a solid understanding of network basics and basic principles of information security.

CO2: Students will be familiarized with various types of cyber attacks, such as malware, socialengineering, and denial-of-service (DoS), as well as common vulnerabilities like SQL injection and cross-site scripting (XSS).

CO3: Students will be able to identify and assess vulnerabilities in computer systems, networks and applications through reconnaissance techniques, vulnerability scanning, and analysis.

DRAFT

CO4: Students will develop practical skills in exploiting security weaknesses within legal and ethical boundaries.

CO5:Students will develop practical skills in using a variety of tools and techniques employed by ethical hackers.

Pre-requisites:

Basic Knowledge of Computers and networking fundamentals

CO/PO Mapping

CO / PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	3	2	2	2	2	3
CO2	3	3	3	2	3	2	3
CO3	3	3	2	2	2	2	3
CO4	3	3	2	2	2	1	3
CO5	3	3	2	2	2	1	3

Legend: 3-High Correlation, 2-Medium Correlation, 1-Low Correlation

Instructional Strategy:

- Provide students with hands-on experience in simulated environments where they can practice hacking techniques ethically.
- Integrate case studies and real-life scenarios to illustrate ethical dilemmas, ethical hacking methodologies, and the consequences of unethical behavior.
- Implement regular quizzes, and practical exercises to evaluate students' understanding of ethical hacking concepts, tools and techniques.
- Throughout the course, a theory-demonstrate-practice-activity strategy may be used to ensure that learning is outcome and employability-based.

DRAFT

Assessment Methodology:

	Continuous Assessment (40 marks)			End Semester Examination
	CA1	CA2	CA3	
Mode	Written Test (Unit I & II)	Written Test (Unit III & IV)	Model Theory Exam	Theory Exam
Duration	2 hours	2 hours	3 hours	3 hours
Exam Marks	60	60	100	100
Converted to	20	20	20	60
Final Marks	40			60

Note:

- CA1 and CA2 Assessment tests should be conducted. CA1 covers unit 1 and unit 2. CA2 covers unit 3 and unit 4. **Best of one** will be considered for 20 Marks for internal assessment.
- CA3 Model examination should be conducted as per the end semester question pattern after completion of all the portions. The marks awarded shall be converted to 20 Marks.

DRAFT

1052235233	Ethical Hacking	L	T	P	C
Theory		3	0	0	3

Unit I	Introduction to Ethical Hacking	
Introduction to ethical hacking: Types of hacking- advantages, disadvantages and purpose of hacking- Types of hackers- Difference between ethical and non-ethical hacking- Ethical Hacking Terminologies- Tools and Skills- Phases of hacking- Laws of the Land. Information Security Overview- CIA triad (Confidentiality, Integrity, Availability)- The Indian IT Act 2000 and Amendments to the Indian IT Act(2008).		09
Unit II	**Reconnaissance & Footprinting**	
Reconnaissance: Active Reconnaissance- Passive Reconnaissance- Footprinting: Domain Name Information- Finding IP Address- Finding Hosting Company- IP Address Ranges- History of the Website.Fingerprinting: Banner Grabbing- application fingerprinting, web application scanning, and DNS fingerprinting. DNS Enumeration.		09
Unit III	**Scanning & Sniffing**	
Scanning: port scanning- Ping Sweep-Scanning Networks- Network discovery- Vulnerability scanning Sniffing: Introduction- Wire trapping and its types, packet sniffing-ARP spoofing, DNS spoofing and MAC flooding, active and passive sniffing,wi-fi sniffing- session hijacking- Man-In The Middle attack, sniffing countermeasures, sniffing detection techniques.		07
Unit IV	**Enumeration, Vulnerabiliy Analysis & Malwares**	
Enumeration- System enumeration- User enumeration- Service enumeration- Vulnerability Analysis- Vulnerability assessment- Common vulnerabilities and exposures (CVE)- Risk assessment. TCP/IP Hijacking- EMAIL Hijacking -Password Hacking- Dictionary Attack-Hybrid Dictionary Attack-Brute-Force Attack-Rainbow Tables- System Hacking- Password cracking- Privilege escalation- Maintaining access. Malware Threats: Types of malware (Types of viruses, worms, trojans, etc.)- Anti-malware tools and techniques.		10

DRAFT

Unit V	Social Engineering & Web Application Security	
Social Engineering: Types of social engineering attacks- Prevention and awareness- Denial of Service (DoS) and Distributed Denial of Service (DDoS) Attacks- DoS and DDoS concepts- DoS and DDoS attack techniques- Mitigation strategies- Web Application Security- Common web vulnerabilities -SQL injection- XSS, CSRF- Introduction to Pen Testing: need for pen testing, types and techniques of pen testing, phases of pen testing.		10
	TOTAL PERIODS	45

Text and Reference Books:

1) David Kim, Michael G. Solomon, "Fundamentals of Information Systems Security", Jones & Bartlett Learning Publishers, 2013

2) Patrick Engebretson, "The Basics of Hacking and Penetration Testing: Ethical Hacking and Penetration Testing Made easy", Elsevier, 2011

3) Kimberly Graves, "CEH Official Certified Ethical hacker Review Guide", Wiley Publishers, 2007 (Unit 3)

4) William Stallings, Lawrie Brown, "Computer Security Principles and Practice", Third Edition, Pearson Education, 2015

5) Allen Harper, Shon Harris, Jonathan Ness, Chris Eagle, Gideon Lenkey, and Terron Williams Grey Hat Hacking: The Ethical Hacker's Handbook, Third Edition, 2011, The McGraw-Hill Companies.

6) AnkitFadia" Ethical Hacking, Second Edition, Macmillan India Ltd, 2006

7) CEH official Certified Ethical Hacking Review Guide, Wiley India Edition, Version 11, 2021.

Web-based/Online Resources:

https://www.udemy.com/topic/ethical-hacking/free/

https://www.coursera.org/learn/ethical-hacking-essentials-ehe

https://www.hackthebox.com/hacker

https://www.cybrary.it/course/ethical-hacking

https://nielit.gov.in/gorakhpur/sites/default/files/Gorakhpur/B01_Ethical_Hacking_220125.pdf

https://osou.ac.in/eresources/DCS-Syllabus-2022-23.pdf

https://www.wscubetech.com/blog/ethical-hacking-course-syllabus/#

Suggested List of Students Activity:

DRAFT

- Virtual environments can be set up to practice hacking techniques in a controlled environment and students can be assigned real-world scenarios where they need to perform penetration tests on simulated corporate networks, web applications, or wireless networks.

- Students can be provided with vulnerable systems to exploit. Reverse engineering techniques can be taught to students by providing them with malware samples or binary executables to analyze.

Continuous Assessment Pattern (CA1 & CA2)

Theory (50)

Allocation of Marks for CA1 & CA2 Theory Tests

Section	Description	Marks	
Part – A	11 questions to be answered out of 12	11 x 2 marks	22
Part – B	4 questions to be answered out of 6	4 x 7 marks	28
TOTAL			**50**

DRAFT

1052235234	Agile Product Development	L	T	P	C
Theory		3	0	0	3

Introduction:

Agile Product Development is a model in Software Engineering, which deals with reliability and quality assurance of the software under development. It provides framework for development of quality software product. The course covers important aspects of product development such as software lifecycle, requirement analysis and documentation, characteristics of good design, design techniques, testing, software implementation, maintenance etc. This course also provides the students with a theoretical understanding of agile software development practices and how small teams can apply them to create high-quality software.

Course Objectives:

The student should be made to

- Define Software Engineering and to understand the phases in a software project
- Understand different software development models.
- Understand the benefits and pitfalls of working in an agile team.
- Understand agile development and testing.
- To learn how the agility is incorporated in Requirement engineering and quality assurance

Course Outcomes:

On successful completion of this course, the student will be able to

CO1: Explain different software development models

CO2: Interpret the concept of agile software engineering and its advantages in software development.

CO3: Analyse the core practices behind the given agile methodologies.

CO4: Interpret how agility is incorporated in Knowledge Management

CO5: Explain and Make use of various tools available to agile teams to facilitate the project and to perform quality assurance in agile team

Pre-requisites: Nil

DRAFT

CO/PO Mapping

CO / PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	2	2			1	1
CO2	3	2	2	1		2	2
CO3	3	3	3	1		2	2
CO4	3	3	3	1		2	1
CO5	3	3	2	2		1	1

Legend: 3-High Correlation, 2-Medium Correlation, 1-Low Correlation

Instructional Strategy:

- **Engage and Motivate:** Instructors should actively engage students to boost their learning confidence.
- **Real-World Relevance:** Incorporate relatable, real-life examples and engineering applications to help students understand and appreciate course concepts.
- **Interactive Learning**: Utilize demonstrations and plan interactive student activities for an engaging learning experience.
- **Application-Based Learning**: Employ a theory-demonstrate-practice-activity strategy throughout the course to ensure outcome-driven learning and employability.
- **Encourage Critical Analysis:** Foster an environment where students can honestly assess experiment outcomes and analyze potential sources of error in case of discrepancies.

DRAFT

Assessment Methodology:

	Continuous Assessment (40 marks)			End Semester Examination (60 marks)
	CA1	CA2	CA3	
Mode	Written Test (Unit I & II)	Written Test (Unit III & IV)	Model Theory Exam	Theory Exam
Duration	2 Hours	2 Hours	3 Hours	3 Hours
Exam Marks	50	50	100	100
Converted to	20	20	20	60
Final Marks	40			60

Note:

- CA1 and CA2 Assessment tests should be conducted.CA1 covers unit 1 and unit 2, CA2 covers unit 3 and unit 4. Best of one will be considered for 20 Marks for internal assessment.
- CA3 Model examination should be conducted as per the end semester question pattern after completion of all the portions. The marks awarded shall be converted to 20 Marks.

DRAFT

1052235234	Agile Product Development	L	T	P	C
Theory		3	0	0	3

Unit I	INTRODUCTION TO SOFTWARE ENGINEERING	
Basics of Software Engineering : Need for Software Engineering – Definition – Software Characteristics –Program versus Software Products- Software Development Life Cycle Models: Introduction –Waterfall Model – Prototyping model – Spiral Model – Iterative Enhancement model – Agile model.		9
Unit II	**AGILE METHODOLOGY**	
Agile Software Development – Traditional Model vs. Agile Model - Classification of Agile Methods – Agile Manifesto and Principles – Agile Project Management – Agile Team Interactions – Ethics in Agile Teams - Agile Documentations – Agile Drivers, Capabilities and Values.		9
Unit III	**AGILE PROCESSES**	
Lean Production – SCRUM- Crystal -Feature Driven Development- Adaptive Software Development - Extreme Programming: Method Overview – Lifecycle – Work Products, Roles and Practices.		9
Unit IV	**AGILITY IN KNOWLEDGE MANAGEMENT**	
Agile Information Systems – Agile Decision Making - Earl_S Schools of KM – Institutional Knowledge Evolution Cycle: Development, Acquisition, Refinement, Distribution, Deployment, Leveraging – KM in Software Engineering – Story Card Maturity Model (SMM).		9
Unit V	**AGILITY IN REQUIREMENTS ENGINEERING & QUALITY ASSURANCE**	
Impact of Agile Processes in Requirements Engineering(RE)– Overview of RE Using Agile – Managing Unstable Requirements – Requirements Elicitation –– Requirements Management in Agile Environment- Agile Requirements Prioritization.– Agile Metrics –- Agility in Quality Assurance.		9
	TOTAL PERIO DS	**45**

DRAFT

Suggested List of Students Activity:

- Presentation/Seminars by students on any recent technological developments in Software Development.
- Blended learning activities to explore the recent trends and developments in the field.
- Roleplay and case studies

Text Books:

1. Roger S. Pressman, Software Engineering – A Practitioner's Approach, Seventh Edition, McGrawHill International Edition, 2010
2. Ken Schawber, Mike Beedle, "Agile Software Development with Scrum", International Edition, Pearson.
3. Robert C. Martin, "Agile Software Development, Principles, Patterns and Practices", First International Edition, Prentice Hall,2014

Reference Books:

1. Lisa Crispin, Janet Gregory, "Agile Testing: A Practical Guide for Testers and Agile Teams", Internationaledition, Addison Wesley.
2. Alistair Cockburn, "Agile Software Development: The Cooperative Game", 2nd Edition, Addison-Wesley
3. Pedro M. Santos, Marco Consolaro, and Alessandro Di Gioia, "Agile Technical Practices Distilled: A learning journey in technical practices and principles of software design", First edition, Packt Publisher.
4. David J. Anderson, Eli Schragenheim "Agile Management for Software Engineering: Applying the Theory of Constraints for Business Results" September 2003,Publisher(s): Pearson,ISBN: 0131424602

Web-based/Online Resources:

1. "The Complete Guide to Agile Software Development" https://clearbridgemobile.com/complete-guideagile-software-development/
2. "Agile Fundamentals Ebook: A Complete Guide for Beginners", https://agileken.com/agilefundamentals-ebook/
3. "Agile Software Development, Principles, Patterns and Practices", First International Edition, Prentice Hall EBook

DRAFT

https://dl.ebooksworld.ir/motoman/Pearson.Agile.Software.Development.Principles.Patterns.and.Practices.www.EBooksWorld.ir.pdf

4. "Agile Software Development",
   https://www.edx.org/course/agile-software-development
5. "Agile Software Development",
   https://www.coursera.org/learn/agile-software-development

Continuous Assessment Pattern (CA1 & CA2)

Theory (50)

Allocation of Marks for CA1 & CA2 Theory Tests

Section	Description	Marks	
Part – A	11 questions to be answered out of 12	11 x 2 marks	22
Part – B	4 questions to be answered out of 6	4 x 7 marks	28
TOTAL			**50**

DRAFT

1052235235	Artificial Intelligence	L	T	P	C
Theory		3	0	0	3

Introduction

Artificial Intelligence has grown to be very popular in today's world. The amount of data that is generated, by both humans and machines, far outpaces humans' ability to absorb, interpret, and make complex decisions based on that data. Artificial intelligence forms the basis for all computer learning and is the future of all complex decision making. Computers are extremely efficient at calculating these combinations and permutations to arrive at the best decision. Artificial intelligence and its logical evolution of machine learning are the foundational future of business decision making.

Course Objectives

On completion of the syllabus contents, the students must be able to

- Describe and use the basic concepts of intelligent agents.
- Design a knowledge-based system.
- Develop general-purpose problem-solving agents, logical reasoning agents, and agents that reason under uncertainty.
- Identify systems with Artificial Intelligence.
- Choose appropriate algorithms for solving given AI problems.
- Design and implement logical reasoning agents.
- Design and implement agents that can reason under uncertainty.
- Apply Artificial Intelligence techniques for problem solving.

Course Outcomes

After successful completion of this course, the students should be able to

CO1: Remember the basic concepts of Artificial intelligence.

CO2: Solve basic AI based problems.

CO3: Define the concept of Artificial Intelligence.

CO4: Apply AI techniques to real-world problems to develop intelligent Systems.

CO5: Select appropriately from a range of techniques when implementing Intelligent systems.

DRAFT

Pre-requisites: Nil

CO/PO Mapping

CO / PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	3	3	1	1	3	1
CO2	3	3	3	1	1	3	2
CO3	3	3	3	1	1	3	1
CO4	3	3	3	1	1	3	2
CO5	3	3	3	1	1	3	2

Legend: 3-High Correlation, 2-Medium Correlation, 1-Low Correlation

Instructional Strategy

- Engage and Motivate: Instructors should actively engage students to boost their learning confidence.
- Real-World Relevance: Incorporate relatable, real-life examples and engineering applications to help students understand and appreciate course concepts.
- Interactive Learning: Utilize demonstrations and plan interactive student activities for an engaging learning experience.
- Application-Based Learning: Employ a theory-demonstrate-practice-activity strategy throughout the course to ensure outcome-driven learning and employability.
- Simulation and Real-World Practice: Conduct demonstrations and hands-on activities in a simulated environment, transitioning to real-world scenarios when possible.
- Encourage Critical Analysis: Foster an environment where students can honestly assess experiment outcomes and analyze potential sources of error in case of discrepancies.

DRAFT

Assessment Methodology

	Continuous Assessment (40 marks)			End Semester Examination (60 marks)
	CA1	CA2	CA3	
Mode	Written Test (Unit I & II)	Written Test (Unit III & IV)	Model Examination	Written Examination
Duration	2 hours	2 hours	3 hours	3 hours
Exam Marks	60	60	100	100
Converted to	20	20	20	60
Marks	20		20	60

- CA1 and CA2 Assessment test should be conducted. Best of one will be considered for the internal assessment of 20 Marks.
- CA3 Model examination should be conducted as per the end semester question pattern. The marks should be converted to 20 marks for the internal assessment.

DRAFT

DRAFT

1052235235	Artificial Intelligence	L	T	P	C
Theory		3	0	0	3

Unit I	INTRODUCTION TO ARTIFICIAL INTELLIGENCE	9

Introduction to AI – Definition - Goals of AI – Applications of AI - History of AI -Types of AI – Components of AI - Human vs Machine Intelligence

Agents – Rationality – Structure of Agents - Problem Solving Agents–Types of Agents - Environments-Nature of Environments

Unit II	PROBLEM SOLVING USING SEARCH TEHNIQUES	9

Search Algorithms: Evaluating Search Strategies - Breadth-first search, Uniform cost search, Depth-first search, Bidirectional Search.

Heuristic Search Strategies: Best First Search, Heuristic Search, A* Search

Unit III	KNOWLEDGE REPRESENTATION AND OPTIMIZATION TECHNIQUES	9

Knowledge Representation: Knowledge-Based Agents, Logic, Propositional Logic: A Very Simple Logic, Ontological Engineering, Categories and Objects, Events, Mental Events and Mental Objects, Reasoning Systems for Categories

Optimization Algorithms: Hill Climbing, Local Beam Search and Genetic Algorithm

Unit IV	GAME PLAYING AND CONSTRAINT SATISFACTION PROBLEMS	9

Game Theory: The Mini-Max search – Alpha-Beta Search- Introduction to CSPs Constraint Networks - Binary and Non-Binary Constraints –Constraint Propagation –Backtracking Search for CSP–Local Search for CSP–Structure of CSP.

Unit V	INTELLIGENT AGENTS	9

Knowledge-Based Agents –Propositional Logic – Propositional Theorem Proving – Propositional Model Checking – Agents Based on Propositional Logic.

Architecture for Intelligent Agents—Agent communication— Argumentation among Agents.

TOTAL PERIODS	45

Suggested List of Students Activity

- Presentation/Seminars by students on any recent technological developments based on the course.
- Periodic class Assessments conducted based on the course.
- Blended learning activities to explore the recent trends and developments in the field.

Text Books and References

1. Artificial intelligence, A modern Approach, 2nded, Stuart Russel, Peter Norvig, Prentice

DRAFT

Hall

2. Artificial Intelligence- 3rd Edition, Rich, Kevin Knight, Shiv Shankar B Nair, TMH

3. Introduction to Artificial Intelligence and Expert Systems, 1st Edition, Patterson, Pearson India, 2015

4. Artificial intelligence, structures and Strategies for Complex problem solving, 5th Edition, George F Lugar, PEA

5. Introduction to Artificial Intelligence, Ertel, Wolf Gang, Springer, 2017

6. Artificial Intelligence, SarojKaushik, 1st Edition, CENGAGE Learning, 2011

Web-based/Online Resources

AI in healthcare

https://www.ibm.com/blog/the-benefits-of-ai-in-healthcare/

Reinforcement Learning

https://en.wikipedia.org/wiki/Reinforcement_learning

Computer vision

https://www.javatpoint.com/computer-vision

Suggested List of Students Activity

- Presentation/Seminars by students on any recent technological developments based on the course.
- Periodic class Assessments conducted based on the course.
- Blended learning activities to explore the recent trends and developments in the field.

Continuous Assessment Pattern (CA1 & CA2)

Theory (50)

Allocation of Marks for CA1 & CA2 Theory Tests

Section	Description	Marks	
Part – A	11 questions to be answered out of 12	11 x 2 marks	22
Part – B	4 questions to be answered out of 6	4 x 7 marks	28
TOTAL			**50**

DRAFT

DRAFT

1052235320	Internet of Things and Digital Twins	L	T	P	C
Practical		0	0	4	2

Rationale

This course will give a hands-on experience to the students in designing and developing Internet of Things applications and models.

Course Objectives

The objective of this course is to

6. Enable the students to understand the basic concepts of Internet of Things.
7. Summarize the functionalities of sensors and actuators.
8. Facilitate the students to design simple IoT concepts.
9. Illustrate the usage of cloud in IoT applications.
10. Introduce digital-twin technology to the students.

Course Outcomes

After successful completion of this course, the students should be able to

CO1: Interface sensors with micro controllers.

CO2: Design Internet of Things models using sensors and actuators.

CO3: Setup a cloud interface to visualize the data.

CO4: Understand the concept of digital twin technology.

CO5: Develop digital twin models using the software.

Pre-requisites

The student should have taken up Computer Networks .

CO/PO Mapping

CO / PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	3	2	1	1	1	3
CO2	3	2	3	1	1	1	3
CO3	3	3	2	1	1	1	3
CO4	3	2	2	1	1	1	3
CO5	3	2	3	1	1	1	3

DRAFT

Legend: 3-High Correlation, 2-Medium Correlation, 1-Low Correlation

Instructional Strategy

- It is advised that teachers make the learning experience more engaging by introducing innovative and interesting ways of teaching.
- The teachers need to expose the students to material in multiple modes help them learn it faster and retain it longer.
- The teacher can focus the pupils' attention on the relevant facts and introduce scientific principles and concepts with the help of demonstration.

Assessment Methodology

	Continuous Assessment (40 marks)			End Semester Examination (60 marks)
	CA1	CA2	CA3	
Mode	Practical Test First 50% Exercises	Practical Test Next 50% Exercises	Model Practical Examination	Practical Exam
Duration	2 hours	2 hours	3 hours	3 hours
Exam Marks	50	50	100	100
Converted to	20	20	10	60
Practical log book / Observation Notebook with bonafide.	10			
Final Marks	40			60

Note:

- Both CA1 and CA2 Assessment test should be conducted. Best of CA1 and CA2 should be considered for the internal assessment of 20 Marks.
- Continuous Assessment Model Practical examination (CA3) should be conducted as per the end semester question pattern after completion of all the portions. The marks awarded shall be converted to 10 Marks.

DRAFT

- The student should submit the duly verified practical log/observation book during CA3..
- Practical logbook / Observation Notebook with Bonafide Carries 10 Marks.

DRAFT

1052235320	Internet of Things and Digital Twins	L	T	P	C
Practical		0	0	4	2

Part - A	**INTRODUCTION TO INTERNET OF THINGS**	
Internet of Things – Sensors – Actuators – Micro Controllers - Introduction to Arduino Board and Arduino IDE – Arduino Programming.		30
Ex.No	Name of the Experiment	
1	Creating different led patterns and controlling them using push button switches.	
2	Controlling servo motor with the help of joystick.	
3	Calculate the distance to an object with the help of an ultrasonic sensor and display it on an LCD.	
4	Basic Burglar alert security system with the help of PIR sensor and Buzzer	
5	Automated LED light control based on input from IR sensor and LDR.	
Part - B	**CLOUD AND DIGITAL TWINS**	
Thing Speak Cloud – Introduction to Digital Twin Technology - Setting up a cloud account in Thing Speak cloud platform.		30
Ex.No	Name of the Experiment	
6	Upload Temperature data to Thing Speak with Node MCU / Raspberry Pi.	
7	Automatic streetlight simulation with Wokwi and Thing Speak.	
8	Create your first thing using ditto.	
9	Query an existing thing using ditto.	
10	Connect an Arduino based device to Eclipse ditto.	
TOTAL HOURS		**60**

DRAFT

Suggested List of Students Activity

- Conduct of Ideathon to generate innovative solutions for real life problems.
- Micro project that shall be an extension of any practical lab exercise to real-world application

Reference

- Internet of Things - A Hands-On Approach, by ArsheepBahga, Vijay Madisetti, Universities Press, First Edition, 2015.
- Internet of Things, Raj Kamal, McGraw Hill Education; First edition, 2017, ISBN 978-9352605224
- Designing the Internet of Things, Adrian McEwen, Hakim Cassimally, Wiley, 1st edition, 2013.
- Digital Twin Technology: Fundamentals and Applications, ManishaVohra, Wiley Online, 2022, Print ISBN:9781119842200.
- Digital Twin Technology, GopalChaudhary, ManjuKhari, Mohamed Elhoseny, 1st Edition, CRC Press, 2022.

Web-based/Online Resources

- Arduino IDE: https://www.arduino.cc/en/IoT/HomePage
- Wokwi Simulator: https://wokwi.com/
- Eclipse Ditto: https://eclipse.dev/ditto/
- Ditto Example: https://github.com/eclipse-ditto/ditto-examples
- Thing Speak Cloud: https://thingspeak.com/

Equipment / Facilities required to conduct the Practical Course

Software Requirement:

1. Arduino IDE
2. Wokwi Simulator
3. Eclipse Ditto
4. Thing Speak Cloud

Hardware Requirement:

1. Arduino kit - 10 Numbers
2. Node MCU / Raspberry Pi - 10 Numbers

DRAFT

3. LED Lights – 10 Numbers
4. 330K Resistor - 10 Numbers
5. Push Button - 10 Number
6. Servo Motor 5 V DC - 10 Numbers
7. Joystick - 10 Numbers
8. Ultrasonic Sensor - 10 Numbers
9. 16x2 LCD Display - 10 Numbers
10. PIR Sensor - 10 Numbers
11. Buzzer - 10 Numbers
12. IR Sensor - 10 Numbers
13. LDR - 10 Numbers
14. LM35 Temperature Sensor- 10 Numbers
15. 5V DC Relay - 10 Numbers
16. Mini Bread Board - 10 Numbers
17. Jumper Wires
18. Data Cables - 10 Numbers

DRAFT

Scheme of Valuation

End Semester Practical Examination

Section	Description	Marks
1	Aim(05), Procedure for the experiment from Part-A (30)	35
2	Aim(05),(Procedure for the experiment from Part-B (30)	35
3	Execution of the experiment from Part-A or Part-B	25
4	Viva voce	5
TOTAL MARKS		**100**

Allocation of Marks for CA1 & CA2 Practical Tests

Section	Description	Marks
1	Aim & Procedure	35

2	Execution and Result	15
	TOTAL MARKS	**50**

1052235440	**Computer Hardware and Networking**	**L**	**T**	**P**	**C**
PRACTICUM		**1**	**0**	**4**	**3**

Rationale

The course aims at making the students familiar with various parts of computers and know the different types of peripherals. They will learn to assemble and repair desktop PC with all its internal components. Students will able to install different types of operating system and all other application software, customization of OS, updating device driver, setting firewall security, junk file removal, data backup and data recovery techniques. The students will learn to setup and configure networking system using various network devices using crimping, punching, setting IP addressing techniques. They are able to share and control resource and internet connection over network. They learn to secure networking system from different types of attacks.

Course Objectives

The objective of this course is to enable the student to

1. Identify the hardware components, assembling a computer, install and configure peripheral device.
2. Install Windows Server OS
3. Do Network Cabling and IP Configuration, Testing
4. Configure DNS Server & AD
5. Configure Web Server, FTP Server, SMTP Server.

Course Outcome

On completion of the following exercises, the students must be able to

1. Identify the hardware components, assembling a computer, Install and configure Peripheral device.
2. Install Windows Server OS
3. Do Network Cabling and IP Configuration, Testing
4. Configure DNS Server & AD
5. Configure Web Server, FTP Server, SMTP Server.

Pre-requisites: Nil

DRAFT

CO/PO Mapping

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	3	3	3	-	-	1
CO2	3	3	3	3	-	-	1
CO3	3	3	3	3	-	-	1
CO4	3	3	3	3	-	-	1
CO5	3	3	3	3	-	-	1

Legend: 3-High Correlation, 2-Medium Correlation, 1-Low Correlation

Instructional Strategy

●It is advised that teachers take steps to get pupils' attention and boost their learning confidence.

●To help students learn and appreciate numerous concepts and principles in each area, teachers should provide examples from daily life, realistic situations, and real-world engineering and technological applications.

●The demonstration can make the subject exciting and foster in the students a scientific mindset. Student activities should be planned on all the topics.

●Throughout the course, a theory-demonstrate-practice-activity strategy may be used to ensure that learning is outcome- and employability-based.

●All demonstrations/Hand-on practices are under a simulated environment (may be followed by a real environment as far as possible).

DRAFT

Assessment Methodology

	Continuous Assessment (40 marks)					End Semester Examination (60 Marks)
	CA1		CA2		CA3	
Mode	Written Test (First 50% of Theory)	Practical Test (50% of Exercise)	Written Test (Next 50% of Theory)	Practical Test (50% of Exercise)	Model Practical Exam	Practical Exam
Duration	2 hours	2 hours	2 hours	2 hours	3 hours	3 hours
Exam Marks	50	50	50	50	100	100
Converted to	20		20		10	60
Practical log book / Observation Note book with bonafide	10					
Final Marks	40					60

Note:

- Continuous Assessment tests (CA1 and CA2) should be conducted for both Theory and Practical portions. **Best of one** will be considered for 20 marks for internal assessment.
- Continuous Assessment Model practical examination (CA3) should be conducted as per the end semester question pattern after completion of all the portions. The marks awarded shall be converted to 10 Marks.
- Practical logbook / Observation notebook with Bonafide carries 10 Marks.

DRAFT

1052235440	Computer Hardware and Networking	L	T	P	C
PRACTICUM		1	0	4	3

Unit I	**COMPUTER HARDWARE**	
Introduction: Hardware, Software and Firmware. Mother board Components, SMPS: Principles of Operation and block diagram of ATX Power supply, Connector Specifications. Processors, Memory: Introduction-Main Memory, Cache memory Secondary Storage: Hard Disk – SSD- Format, Partition I/O devices and interface Keyboard-Mouse-Printers-Scanner-Displays Graphic Cards		4
Ex.No	Name of the Experiment	
1	Assemble and disassembling a computer & Troubleshooting (Study Experiment). Hard Disk/SSD a) Partition and Format. b) Scan Disk, Disk Cleanup, Disk De-Fragmentation	6
Unit II	**OPERATING SYSTEM & PERIPHERAL DEVICE INSTALLATION**	
Operating System-Server OS Installation, User Creation, Disk Cleaning, Defragmentation .Peripheral device (Scanner, Web cam, and bio-metric) Installation & Updating of Device Driver Software.		2
Ex.No	Name of the Experiment	
2	Install and configure any one device (Printer, Scanner, Web cam, bio-metric device) with system and troubleshoot the problems.	6
3	Windows Server OS installation	6
Unit III	**ADMINISTERING WINDOWS SERVER**	
Active directory- Use of AD-Installation & Configuration of AD, Domain Name Service Use of DNS-Installation & Configuration of DNS. Web Server - Installation & Configuration of IIS web Server –HTTP Protocol Usage. FTP Server - Installation & Configuration of FTP Server-FTP Protocol Usage .SMTP Server - Installation & Configuration of SMTP Server		3

DRAFT

Ex.No	Name of the Experiment	
4	Installation and configuration of DNS Server	6
5	Installation and configuration of DHCP Server	6
6	Installation and Configuration of any one of Service (Telnet, FTP Server, Web Server, SMTP Server)	6
Unit IV	**NETWORK DEVICES**	
Network Devices: Cable, Cable Crimping (Cross, Straight Through Cabling), Switch -Features and concepts of Switches – Routers (Wired and Wireless) – Gateways. IP Addressing: Dotted Decimal Notation – Subnetting & Supernetting, Data backup and data recovery, Firewall, VPN		6
Ex.No	Name of the Experiment	
7	Practice the following cabling works in a network. a) Cable Crimping b) Standard Cabling c) Cross Cabling d) Testing the Crimped cable using a Cable tester	6
8	Create a Network topology using any network simulation software.	6
9	IP Configuration & Testing a) Configure Host IP, Subnet Mask and Default Gateway in a system in LAN (TCP/IP Configuration). b) Configure Internet connection and use IPCONFIG, PING /Tracert and Netstat utilities to Debug the Network issues.	6
10.	Data Backup & Network Security a. Create and configure user accounts (Administrative and Standard) in Windows. b. Create automated backups to ensure no data loss & you always have a recent backup c. Create rules on firewall to allow clients to connect to the Server Service	6

DRAFT

TOTAL PERIODS	**75**

Suggested List of Students Activity

I. PC Assembling, Troubleshooting of Hardware effects using indicators.

ii. OS installation & Configuration in Server and Client

iii. Printer, Scanner, Biometric, Camera Installation and Configuration

iv. Usage of various networking tools

Reference

1. Computer Installation And Servicing D.Balasubramanian Tata Mc-Graw Hill, New Delhi Second Edition 2010
2. Data Communication and Networking, Behrouza.Forouzan, Tata Mc-Graw Hill, New Delhi.
3. Computer Networks, Fifth Edition, Andrew S. Tanenbaum, David J. Wetherall
4. Installation Technician Computing and Peripherals, Class XII NCERT Book For Computer Science

Web-based/Online Resources

1.https://epathshala.nic.in/process.php?id=students&type=eTextbooks&ln=en

2.https://yousharedsite.files.wordpress.com/2018/09/mcsa-windows-server-2012-complete-study-guide-exams-70-410-70-411-70-412-and-70-417.pdf

3.https://rajshaladarpan.nic.in/sd4/home/public2/VocationalSchool/Textbook/Electronics%20&%20Hardware%20(%20Installation%20Technician%20&%20Computing%20Peripherals)%20Class%2011th%20Text%20Book.pdf

Equipment / Facilities required to conduct the Practical Portion

LIST OF EQUIPMENTS

Hardware Requirements:

Desktop Systems, Laser Printer, Web camera, Biometric Device, Scanner, Crimping Tool, Screwdriver set, Network Cables, Switch, Router, Cable, Cable Tester, RJ45

Software Requirements:

Windows server OS, Oracle Virtual Box, Windows OS(Host System),Simulation Software (GNS3/ Cisco Packet Tracer)

Continuous Assessment Pattern (CA1 & CA2)

Theory (50)

Allocation of Marks for CA1 & CA2 Theory Tests

DRAFT

Section	Description	Marks	
Part – A	11 questions to be answered out of 12	11 x 2 marks	22
Part – B	4 questions to be answered out of 6	4 x 7 marks	28
TOTAL			**50**

Practical (50)

Allocation of Marks for CA1 & CA2 Practical Tests

Section	Description	Marks
1	Aim & Program	35
2	Execution and Result	15
TOTAL		**50**

Board Practical Examination

PART – A

1.Hard Disk/SSD

a) Partition and Format.

b) Scan Disk, Disk Cleanup, Disk De-Fragmentation

2.Install and configure any one device (Printer, Scanner, Webcam, Bio-metric device) with system and troubleshoot the problems.

3.Server OS installation

4.Installation and configuration of DNS Server

5.Installation and configuration of DHCP Server

PART – B

6.Installation and Configuration of any one of Service (Telnet, FTP Server, Web Server, SMTP Server)

7.Practice the following cabling works in a network

a) Cable Crimping

b) Standard Cabling

c) Cross Cabling

d) Testing the Crimped cable using a Cable tester

8.Create a Network topology using any network simulation software.

DRAFT

9.IP Configuration & Testing

a) Configure Host IP, Subnet Mask and Default Gateway in a system in LAN (TCP/IP Configuration).

b) Configure Internet connection and use IPCONFIG, PING / Tracert and Netstat utilities to Debug the Network issues.

10. Data Backup & Network Security

d. Create and configure user accounts (Administrative and Standard) in Windows.

e. Create automated backups to ensure no data loss & you always have a recent backup

a. Create rules on firewall to allow clients to connect to the Server Service

SCHEME OF VALUATION		
SNO	**ALLOCATION**	**MARKS**
1	Aim(05) ,Writing procedure from Part – A (30)	35
2	Aim(05), Writing procedure from Part – B (30)	35
3	Executing any one Expeiment (Part A or Part –B)	15
4	Output	10
5	Viva Voce	05
6	Total	100

DRAFT

DRAFT